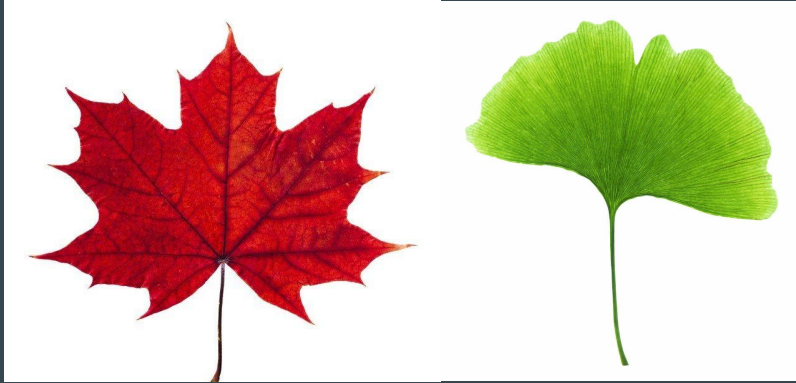


Using Skinning in Simulating Leaf Growth and Development

...

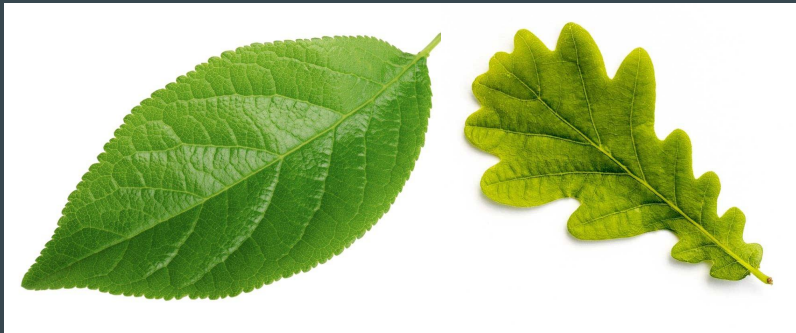
Sua Lim

Introduction: Background



Pixers. (n.d.). Autumn maple leaf isolated on white background [Wall mural image]. Retrieved August 20, 2025, from <https://pixers.ca/wall-murals/autumn-maple-leaf-isolated-on-white-background-nd-46757753>

Gustoimages / Science Photo Library. (n.d.). Ginkgo leaf [Photograph]. Science Photo Gallery. Retrieved August 20, 2025, from <https://sciencephotogallery.com/featured/ginkgo-leaf-gustoimagesscience-photo-library.html>



HarperCollins Publishers. (n.d.). Leaf. In Collins English Dictionary. Retrieved August 20, 2025, from <https://www.collinsdictionary.com/dictionary/english/leaf>

Silvotherapy. (n.d.). The different types of native oak in the UK. Retrieved August 20, 2025, from <https://silvotherapy.co.uk/articles/the-different-types-native-oak-uk>

- Leaves exhibit diverse shapes, shaped by both internal developmental processes and structure, as well as external environmental factors. Leaf outgrowth is primarily governed by three main features:
 - Vascular system: controls direction of growth
 - Margin: controls outline of the leaf, patterning (where is growth promoted/prohibited)
 - Leaf blade: controls size and surface area, expansion between veins
- Specifically, there is an interplay between margin and vascular system: margin patterning (convergence points) defines where new veins initiate from the edge of the leaf. Then, margin patterning guides veins, and veins shape the evolving margin.
- Additionally, leaf shape can be influenced by external environmental factors such as temperature, water availability, light intensity, and nutrient levels, which can modify growth dynamics and developmental outcomes

Introduction: Importance

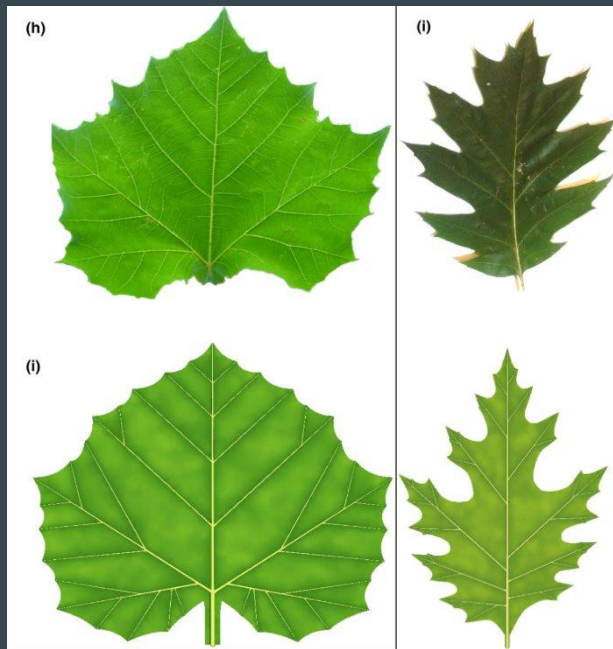
In computer graphics...

- Creating natural leaf shapes that are close to what we see in nature is important in computer graphics context as it allows us to create visually natural outdoor scenes, which in turn helps in creating realistic graphics

In biology...

- Assist biologists model and understand real-world leaf growth patterns:
 - Used to visualize plant development over time, reducing the need for wet-lab experiments
 - By tweaking growth parameters, be able to formulate hypotheses about leaf shape and size
 - Used alongside real data, such simulations could contribute to phenotyping (studying physical traits) in plant biology.
 - Help quantify and model how complex shapes emerge, providing insight for computational biology and biomimetics

Introduction: Previously

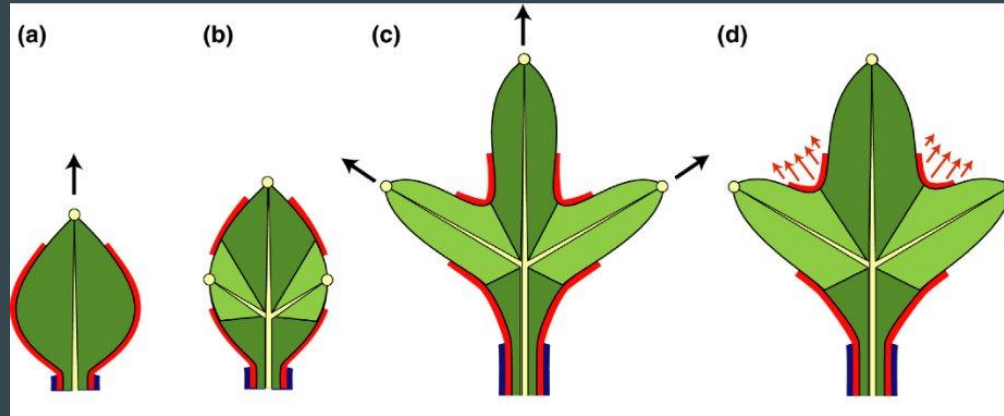


Runions, A., Lane, B., & Prusinkiewicz, P. (2017). A common developmental program can produce diverse leaf shapes. *New Phytologist*, 216(2), 401–418. <https://doi.org/10.1111/nph.14449>

Runions, A., Lane, B., & Prusinkiewicz, P. (2017). A common developmental program can produce diverse leaf shapes. *New Phytologist*, 216(2), 401–418. <https://doi.org/10.1111/nph.14449>

- Due to the fascinating characteristics of leaf growth and development, there has been previous studies in developing leaf growth simulation model that closely aligns with how leaf grows biologically
- Here, we consider the model proposed in the study by *Runions et al. (2017)*, titled ‘A common developmental program can produce diverse leaf shapes.’

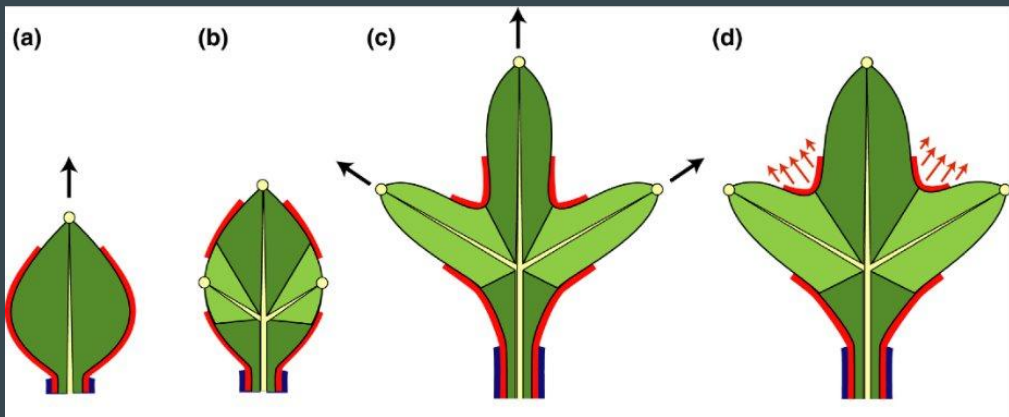
Introduction: Previously



The above figure shows how a leaf grows step by step in the model considered. At first, the edge of the leaf has special signals (shown in different colors) that guide its shape. One signal (red) helps create dips and curves in the edge, while another (blue) marks the stem area.

- Starting out, the leaf has a basic shape.
- The main central vein grows, which pulls the leaf outward and helps it expand. New convergence points (yellow dots) appear along the edge, where new side veins grow. These veins divide the leaf into different sections.
- In the next step, these veins grow more, and each section of the leaf stretches differently depending on the direction of the veins.
- The leaf edge keeps expanding, filling in the space between veins (called webbing), and this continues shaping the final leaf form.

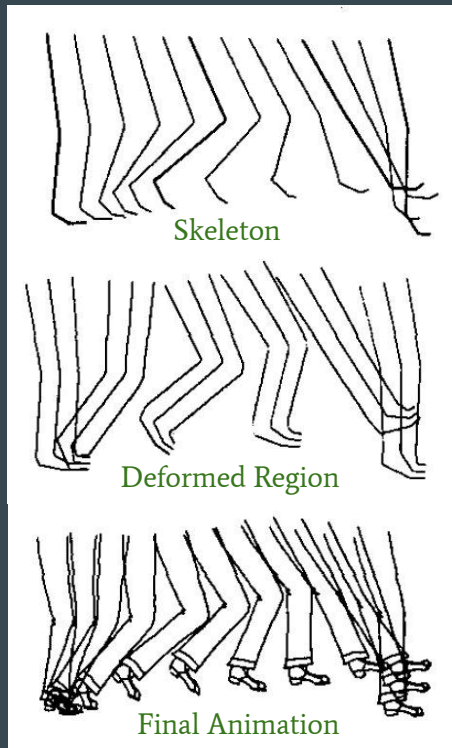
Introduction: Limitation



Runions, A., Lane, B., & Prusinkiewicz, P. (2017). A common developmental program can produce diverse leaf shapes. *New Phytologist*, 216(2), 401–418. <https://doi.org/10.1111/nph.14449>

- However, the model needed to incorporate ad-hoc methods to conceal undesirable artifacts
 - These ad-hoc methods were often not biologically explainable
- For example, fairing had to be used to fill in the discontinuity in the leaf margin as the margin expanded and grew
 - When a segment of the margin is pushed outwards by the extension of a vein, it becomes geometrically disconnected from the adjacent portions of the margin
 - This creates a discontinuity or gap in the overall margin geometry. Hence fairing was acquired to smooth the connection between the displaced segment and the rest of the curve, restoring continuity and preserving a natural, smooth margin

Introduction: Proposed



- Hence, the proposed research introduces an alternative approach to simulating leaf growth and development
- In our context, we use the **skinning** technique to model leaf growth and development
- Skinning is commonly used in animation to deform a character's mesh smoothly based on the movement of an underlying skeleton
- The project defines the relationship between a leaf's margin and its branching veins as contour and skeleton respectively
- Incorporating skinning provides solid mathematical description of associating margin and branches, as well as a geometric interpretation of leaf growth

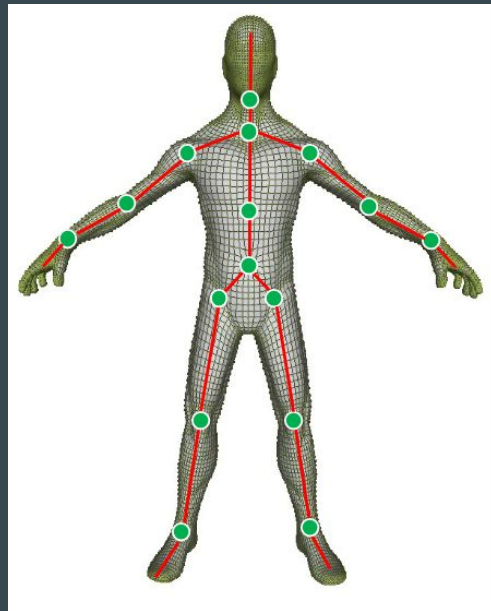
Process/Method

What is Skinning?

- In animation, skinning is the process of binding a 3D model to a skeleton so that the model can move and deform in a realistic way when the skeleton is animated

How it works?

- Model (Mesh): Object made of polygons (vertices, edges, and faces)
- Skeleton (Rig): A hierarchy of bones or joints that control how the mesh moves
- Skinning: The process of assigning weights to each vertex on the mesh, telling it how much it should be influenced by each nearby bone



Process/Method

In our context...

- Leaf veins → skeleton (branching structure)
 - Hierarchical structure of transformations
- Leaf margin → contour (“mesh” in 2D)
 - Smooth curve

Which allows for the margin to deform along with the vein/branch that it is associated with

- Each point on the margin is associated/bound to a particular vein/branch
- Margin point deforms with the branch (transformation) associated with it

New veins/branches are also dynamically added during simulation, where margin points are rebound to preserve the shape and allow for smooth deformation

★ Different from traditional skinning!

Process/Method: Which type of skinning?

There are many skinning techniques used in animation and character deformation, each with different strengths and complexities. In this project, we focus on three different methods:

1. Linear blend skinning: the most common skinning type
2. Skeletal skinning: derived from *Runions et al. (2017)*, 'A common developmental program can produce diverse leaf shapes.'
 - a. Adjusted to fit into the idea of skinning
3. Interpolative skinning: to handle stretching of bones
 - a. For growth simulation
 - b. Idea from Jacobson and Sorkine (2011), 'Stretchable and Twistable Bones for Skeletal Shape Deformation'
 - i. Bone stretching is not handled well in traditional skinning and often introduces artifacts

Process/Method: Which type of skinning?

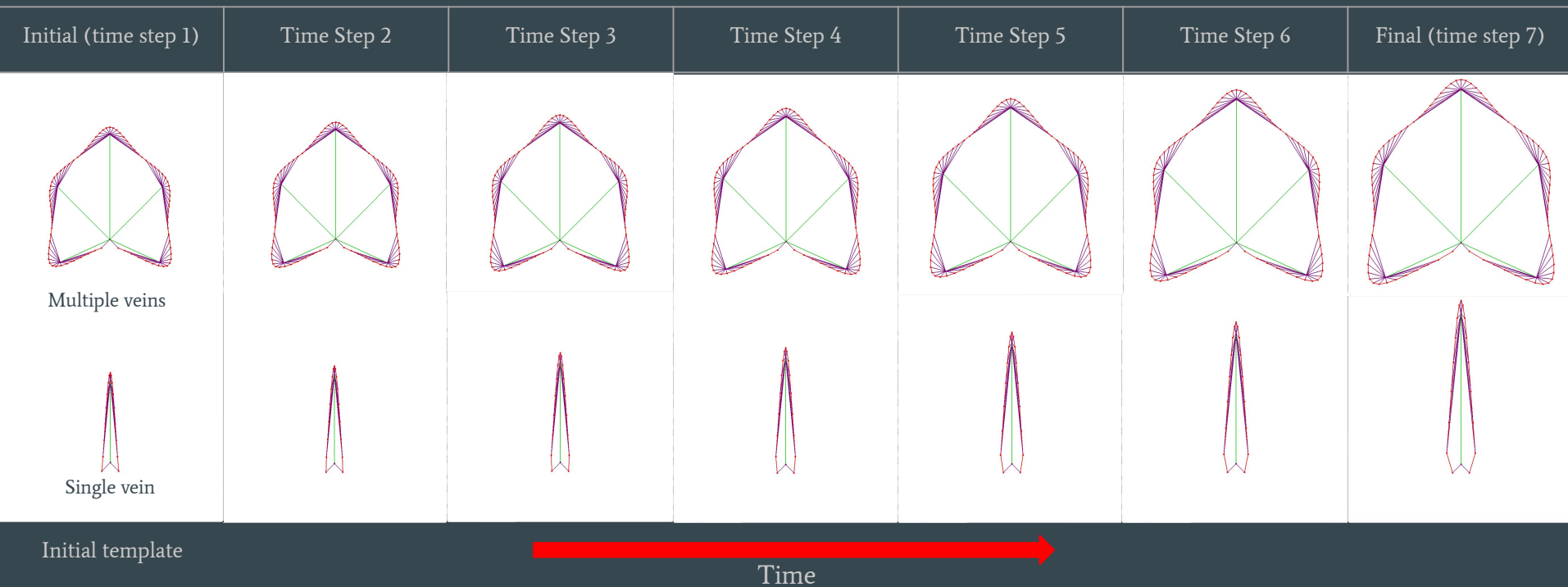
1. Linear Blend Skinning:

- a. Most common skinning technique
- b. Each margin point is influenced by multiple branches/veins
- c. Each influence has a weight (a value between 0 and 1)
 - i. Each margin point has a weight vector
 - ii. For each margin point:
 1. Initialize $W_i = 1$ for closest branch i and the rest 0
 2. Blend the weight with neighboring points (take the repeated average with adjacent points)
- d. Final vertex position = a weighted average of the vertex transformed by each influencing bone

Then: linear blend skinning is a method used in animation where a character's surface moves by smoothly blending the influence of nearby bones

Note: linear blend skinning has a transformation associated with each **branch**

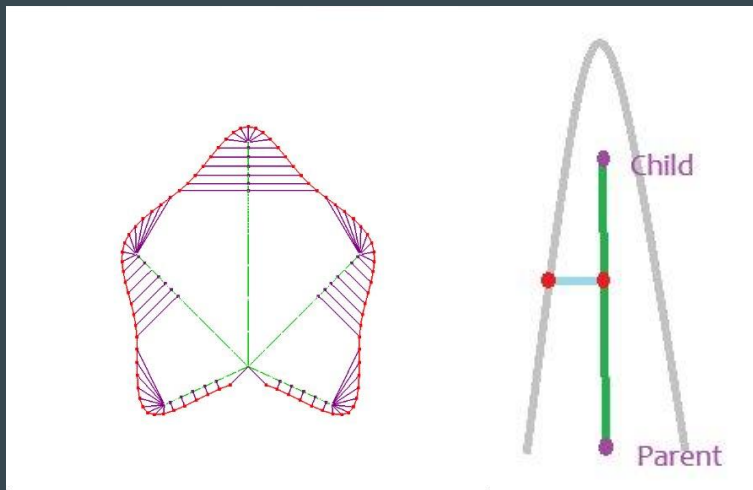
Progression of Method



Note: the binding relationship shows the binding for the closest branch; does not correctly visualize the blended weight/binding ¹²

Process/Method: Which type of skinning?

2. Skeletal Blending:

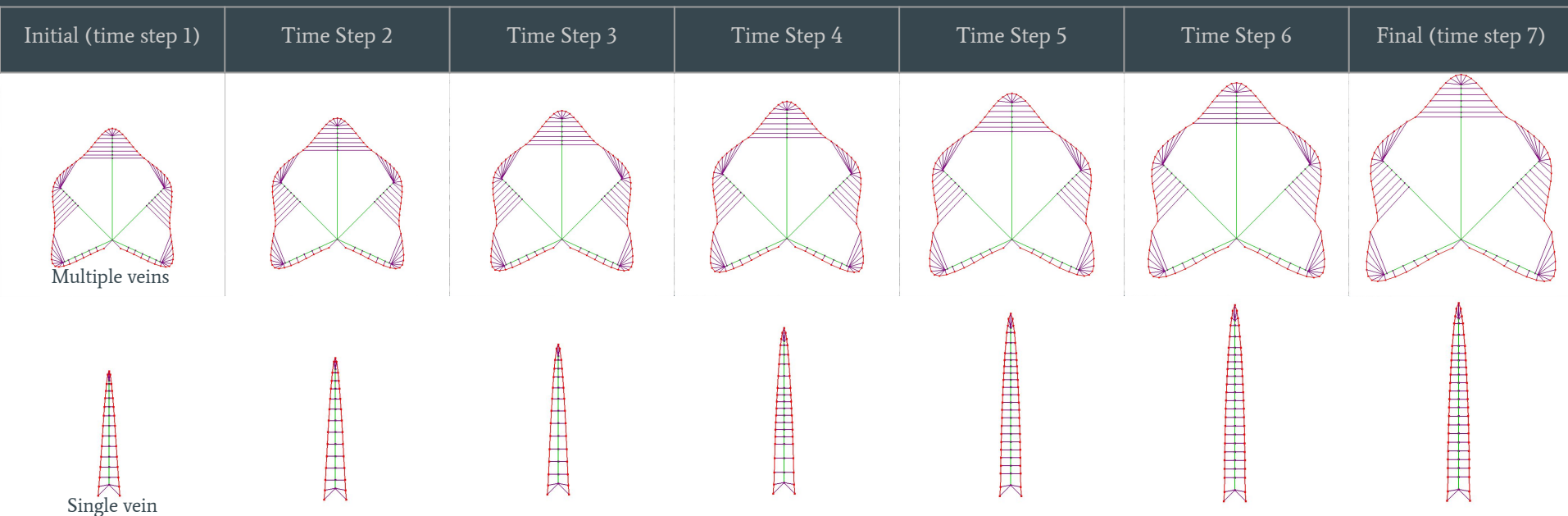


- Green: branching structure
- Red: contour/margin
- Purple: margin point - branch binding

- Based on method incorporate in previous study
- Each margin point is bound to the closest branch
 - Single branch
- Binding happens in the closest point on the branch (specific point on the branch)
- Final transformation for the margin point:
 - Linear interpolation of the transformations associated with the endpoints of the branch
 - Transformation associated with the binding point

Then: skeletal skinning is a method where a character's surface moves by associating to the closest point on the branching structure. Unlike linear blend skinning, skeletal skinning has a transformation associated with each **node of the branch**, and the specific transformation associated to a point on the branch is achieved by linearly interpolating the two nodes (endpoints)

Progression of Method



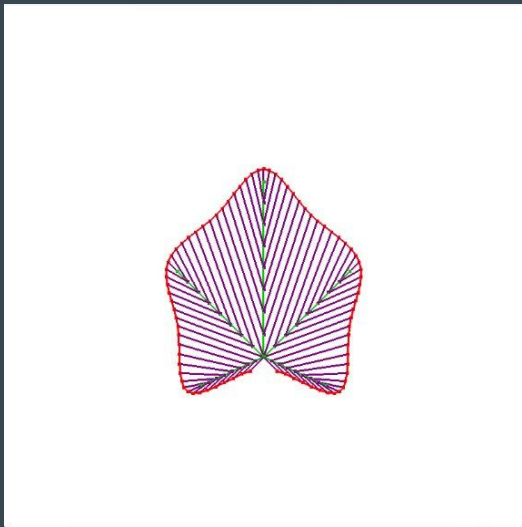
Initial template



Time

Process/Method: Which type of skinning?

3. Interpolative Skinning



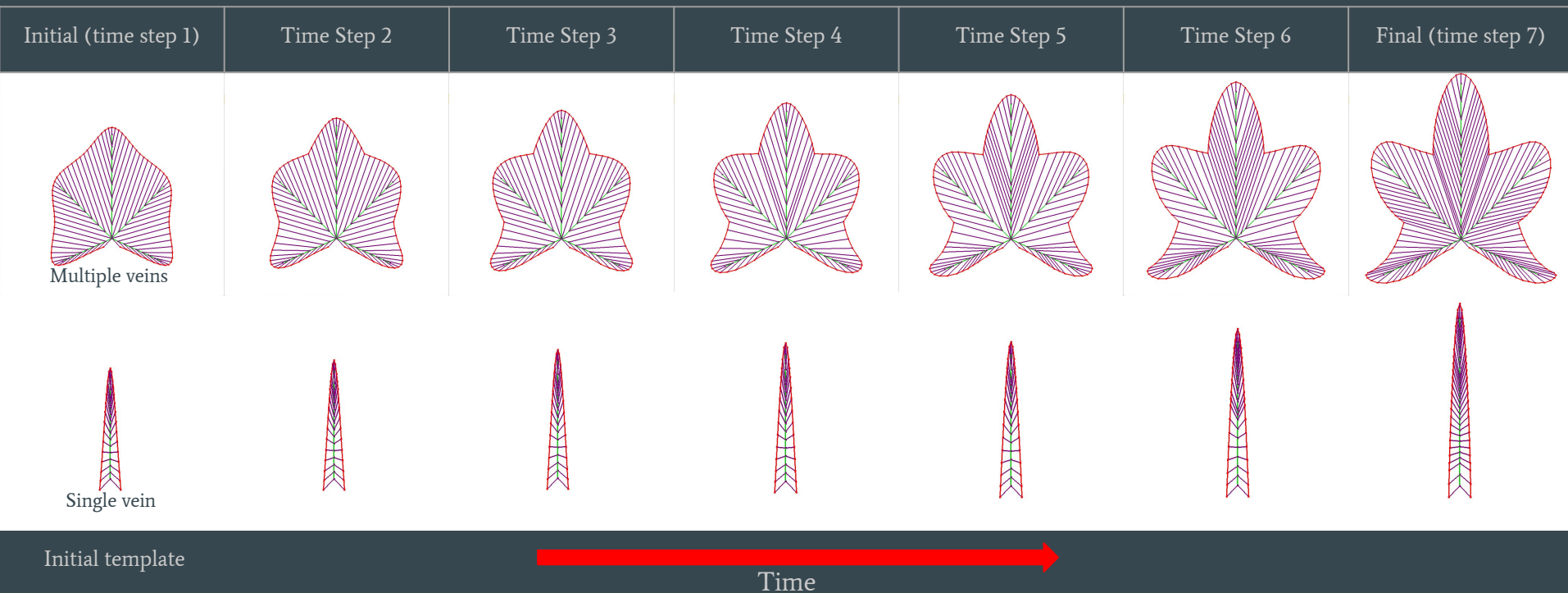
- Each margin point is bound to one branch
- Similar to Skeletal Blending, but instead of taking the closest point as the binding position, linearly interpolate **all** the margin points bound to the **same** branch
- Smooth interpolation removes the issue in Skeletal Blending:
 - Abrupt change in binding between two consecutive margin points results in discontinuity in the contour curve

Then: interpolative skinning is a method where a character's surface moves by first associating to the closest point on the branching structure, then evenly distributing the bindings along the entire structure

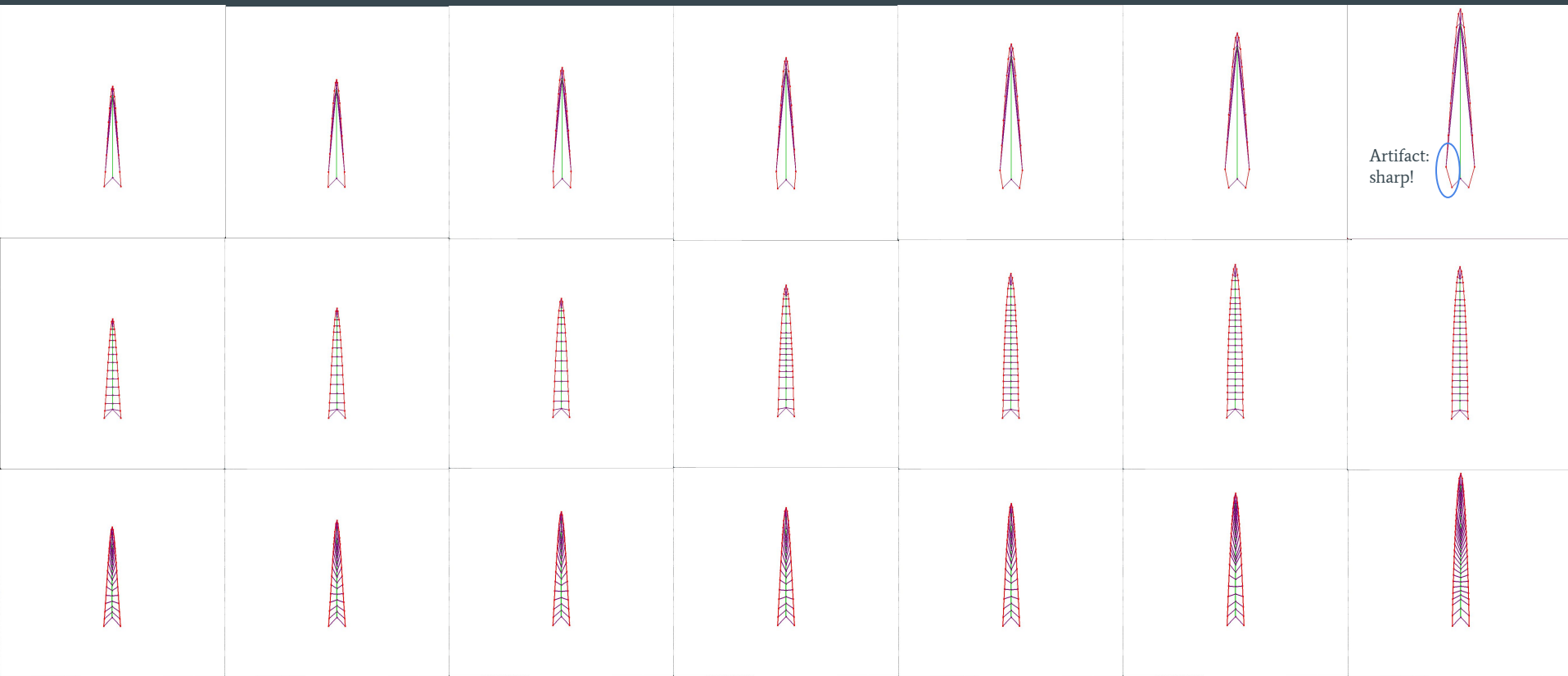
Similar to skeletal skinning, interpolative skinning has transformation associated to each **node of the branch**, and the specific transformation associated to a point on the branch is achieved by linearly interpolating the two nodes (endpoints)

★ Final choice of skinning!

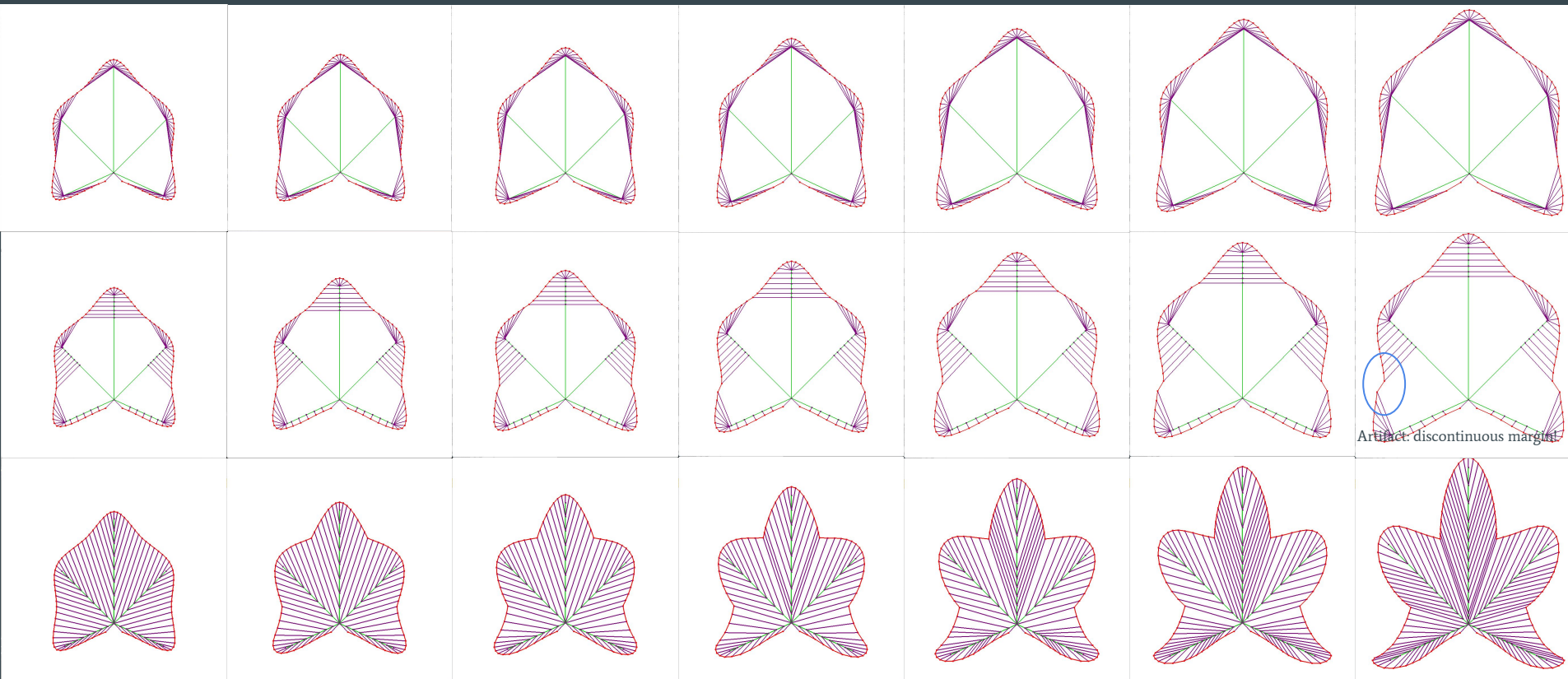
Progression of Method



Comparison of Methods

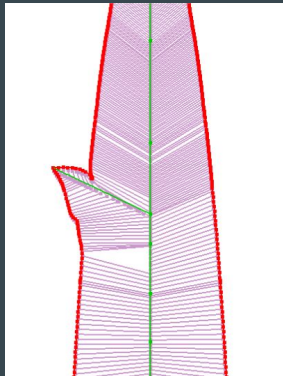
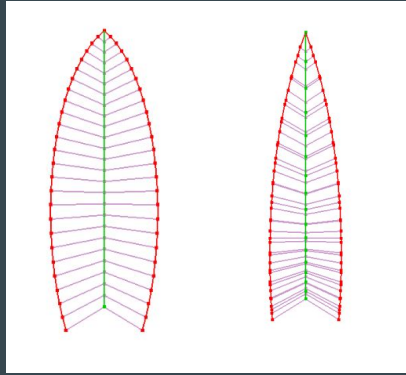


Comparison of Methods

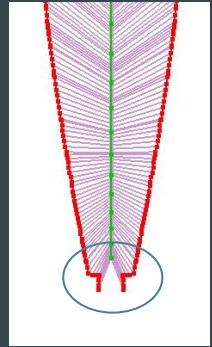


Time

Process/Method: Additional methods



- Petiole →
 - Stalks that attaches a leaf to the stem of a plant
 - Constraints expansion at the base
- Add new margin points:
 - When leaf margin grows along with the veins/branches, the margin gets “stretched” which can result in sharp contour
 - Adding new margin points if the distance between two points become large helps in reducing the sharpness as the margin expands
- Subdivide vein
 - Correct interpretation of how growth rate changes along the branch
 - Growth gradient implementation
- Dynamically add new vein segments
 - Veins are added **during** simulation (not starting with multiple veins)



Comparison of Methods: Why Interpolative Skinning?

Then, comparing the resulting shape from the 3 methods, we can conclude the following:

1. Linear blend skinning provides smooth indentation with multiple blending, but when there is only one vein, there is no blending of weights and results in a sharp margin between point bound to the root and the rest
 2. Skeletal skinning introduces sharpness and/or discontinuity between consecutive margin points that are bound to different veins
 3. Interpolative skinning removes the discontinuity introduced in skeletal skinning by reducing the gap between margin-branch binding, while still preserving indentation control (since there are margin points that are bound near the root/base region)
 - a. Hence, interpolative skinning retains the benefits of the above two while improving their limitations
- ★ Interpolative skinning is the final choice of skinning!

Process/Method: Transformations

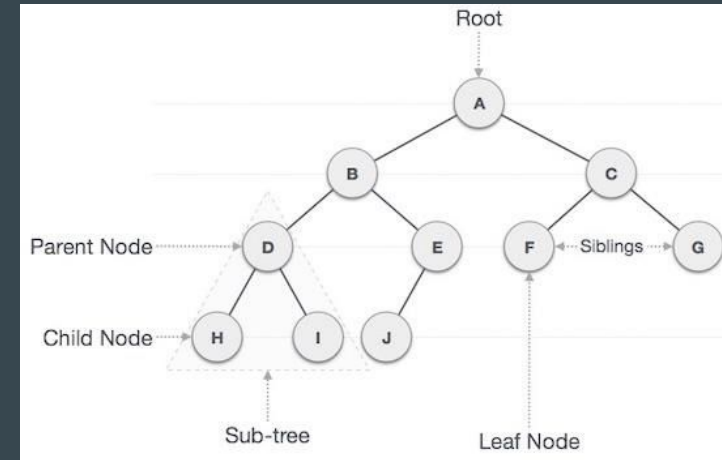
Transformations:

- In skinning, there are two types of transformations:
 - Rest/local pose: initial stationary position
 - Animated/global pose: deformation over time → **responsible for simulating growth!**
 - Final transformation = animated pose * rest pose
- Type of transformation deforms the shape differently
 - Rotation? Scaling (uniform vs non-uniform)? Translation?
- Any linear transformation can be broken down into a combination of the following three: rotation, scaling, translation
- Then, skinning allows us to handle the above transformations appropriately:
 - Translation: translation length is linearly interpolated
 - Scaling: scaling matrix is linearly interpolated
 - Rotation: rotation parameter (angle) is linearly interpolated

Process/Methods: Transformations

In our context...

- Each branch has an associated transformation (combination of rotation, scaling, and translation) that first puts it into initial rest position then deforms over time
- Margin points associated with the branch deforms with the bound branch
 - Associating margin points with a branch = binding
- Since branches are organized in a hierarchical structure, the child branch inherits the parent's transformation
 - Like a tree data structure!

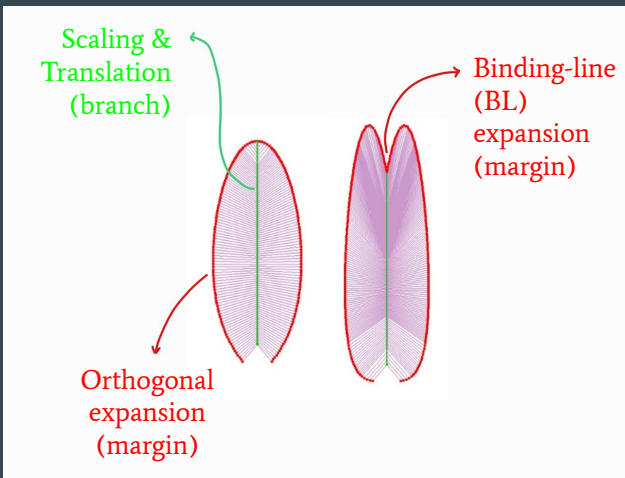


Tutorialspoint. (n.d.). *Tree data structure*.
https://www.tutorialspoint.com/data_structures_algorithms/tree_data_structure.htm
(Accessed August 22, 2025)

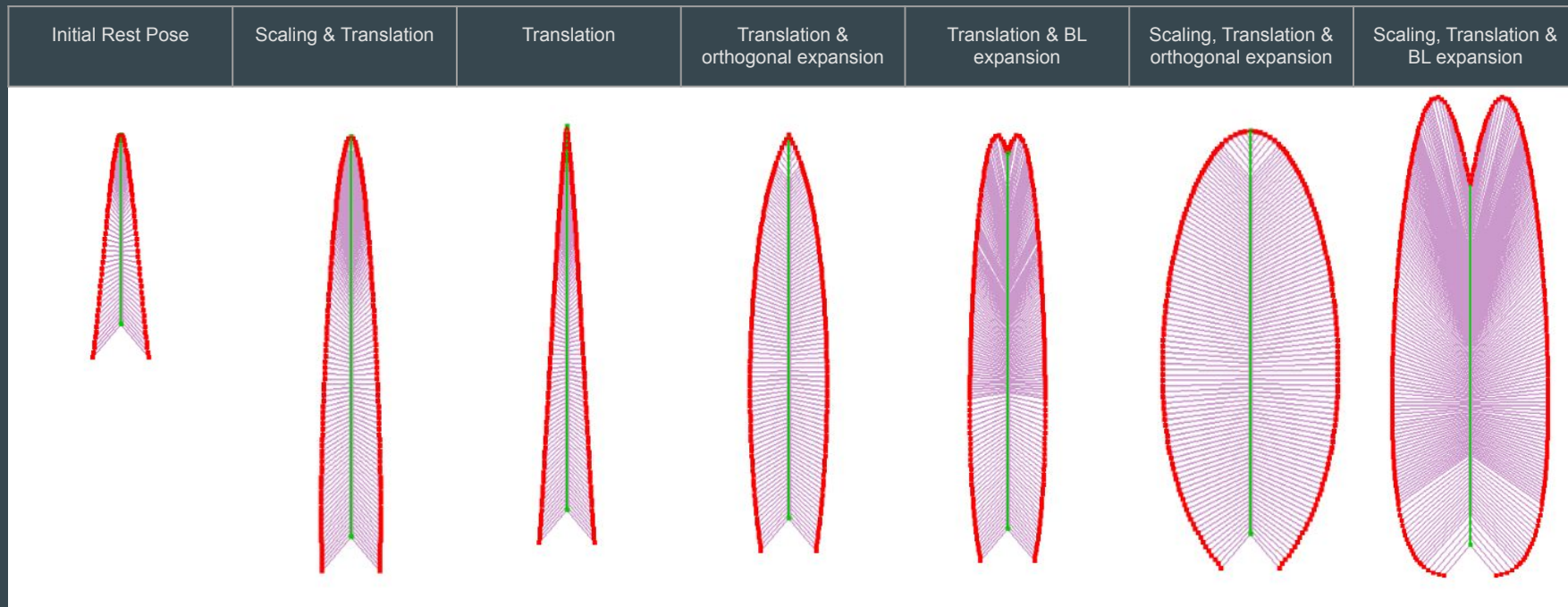
Process/Methods: Transformations

Transformation incorporated in the project:

- Scaling & translation: branch transformation
 - Responsible for branch/vein growth
 - Core skinning part → hierarchical structure of transformation
 - Margin points bound to the branch moves by the associated transformation
 - **Scaling** has two types: local/rest pose scaling and animated/deformation scaling
 - Translation is only local/rest pose
- Orthogonal expansion: margin transformation
 - Non-hierarchical → only associated with a specific margin point
 - Responsible for orthogonal (width) growth
 - Gives spatial area to the leaf blade
- Binding-line expansion (BL expansion): margin transformation
 - Non-hierarchical → only associated with a specific margin point
 - Responsible for leaf tip shape
 - Indented vs flat
 - **BL expansion can be separated into orthogonal and parallel**

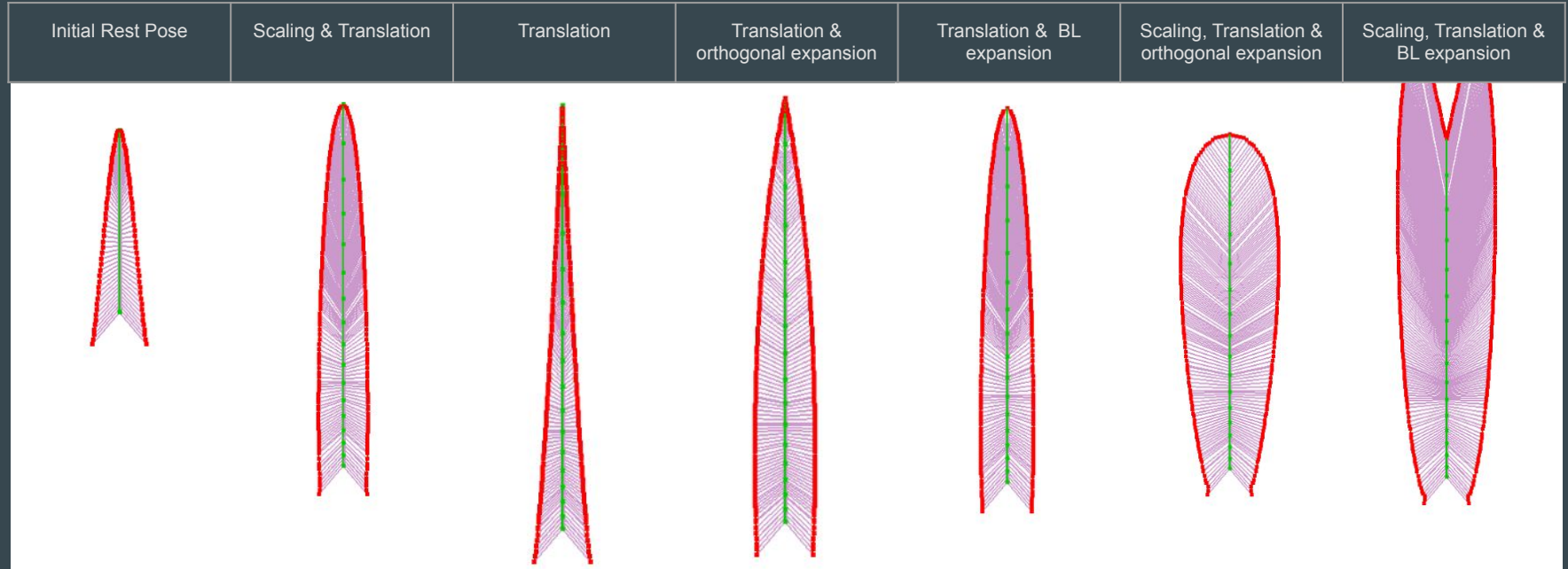


Findings/Results: Transformation Comparison



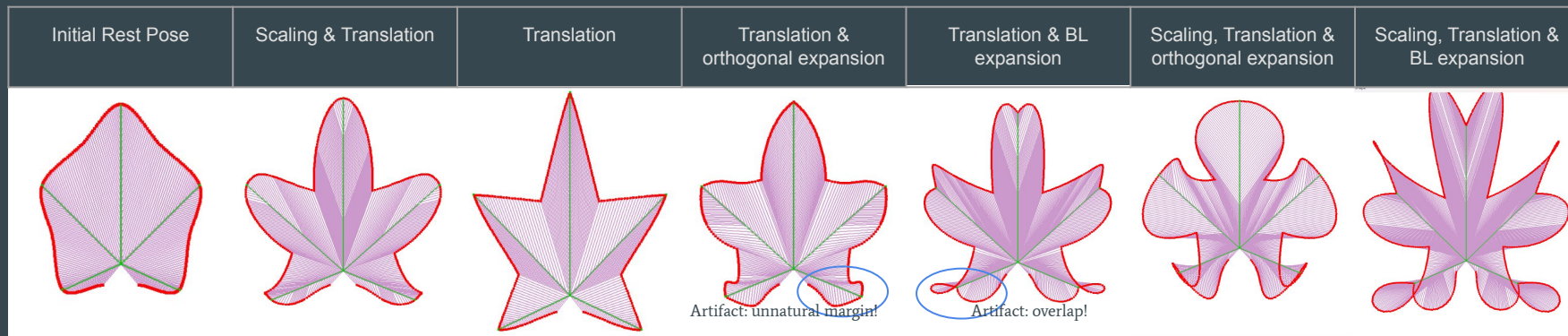
No subdivision of branches: growth rate is approximated by linearly interpolating the branch nodes (endpoints)

Findings/Results: Transformation Comparison

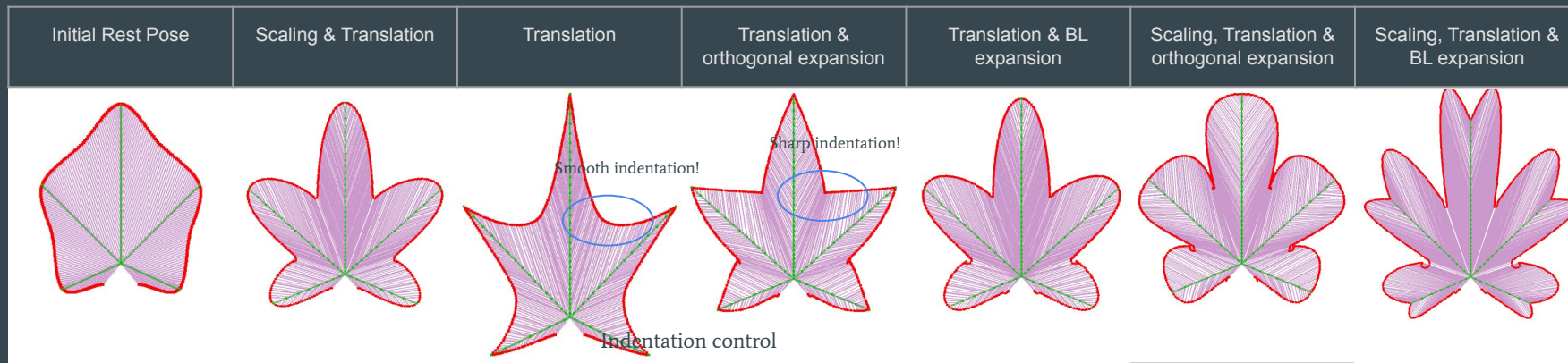


Branch subdivision: growth rate is correctly distributed along the vein, rather than taking an approximation by linear interpolation

Findings/Results: Transformation Comparison



No subdivision of branches



Branch subdivision

Findings/Results: Comments on Effect of Branch Subdivision

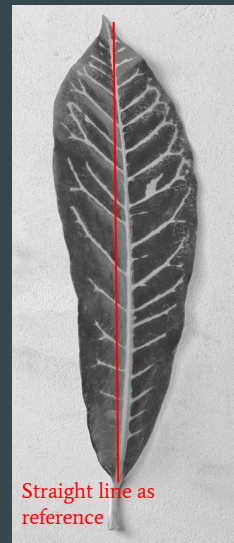
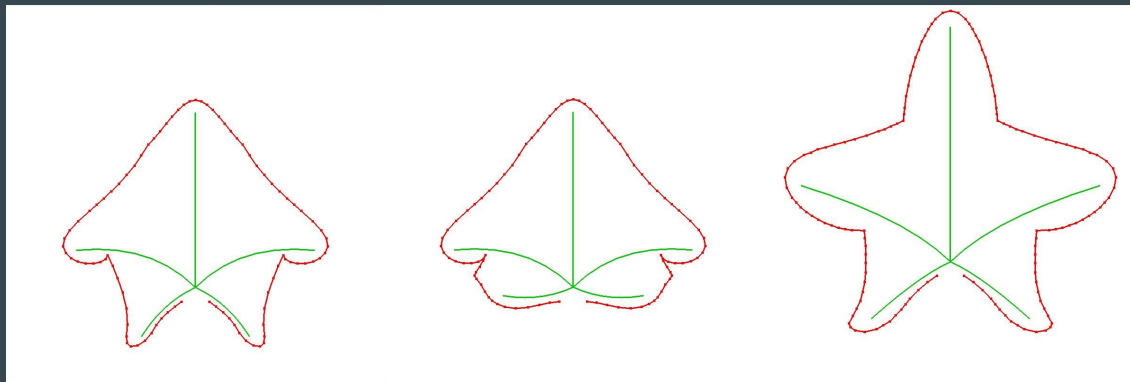
With multiple-vein branching structure, the effects of subdividing branches are clearer:

- With more combinations of transformations, we start to see more artifacts for shapes with no subdivision of branches
 - Margins between veins start to overlap, creating a loop
 - Unnatural bump and bending of margin with differing angle and/or length veins
- However, with subdivision of branches, such artifacts are minimized
 - Because subdivision correctly distributes the growth rate along the vein axis rather than taking a simple approximation
- Also, with subdivision of branches, we can control the shape of the indentation
 - By incorporating different transformation types, we can achieve a smoother or sharper indentation

Findings/Results: Transformation Comparison

Rotation:

- Rotation allows us to capture phenomena such as bending of veins
 - Branch transformation
 - Some types of leaves have curved veins
 - Veins might bend as leaves grow
 - But they are not common in nature



Grant, A. (2021, December 1). Leaf veins are turning yellow – What causes leaves with yellow veins [Image]. Gardening Know How. <https://www.gardeningknowhow.com/houseplants/hpgen/green-leaves-have-yellow-veins.htm> (Accessed August 22, 2025)

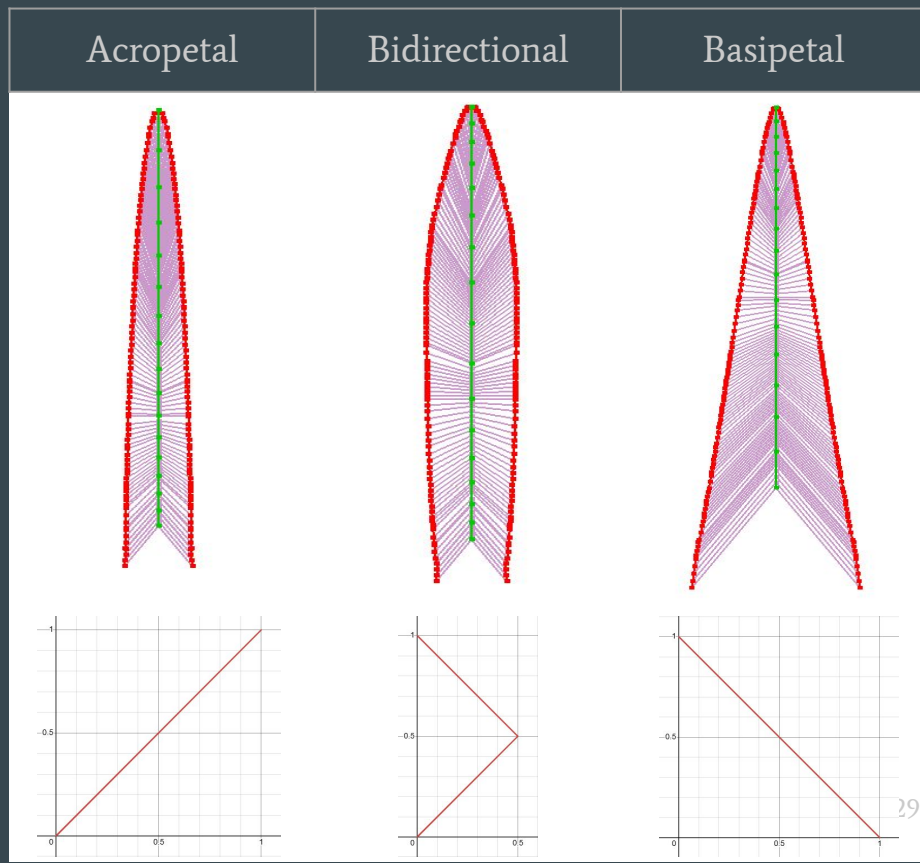
Varying Parameter: Growth Gradient

Growth gradients shape leaves by driving differential growth across regions, resulting in features like curves, lobes, or elongation (depending on where growth is most pronounced). These gradients can either promote or restrict growth along the vein/axis, ultimately influencing the leaf's final form.

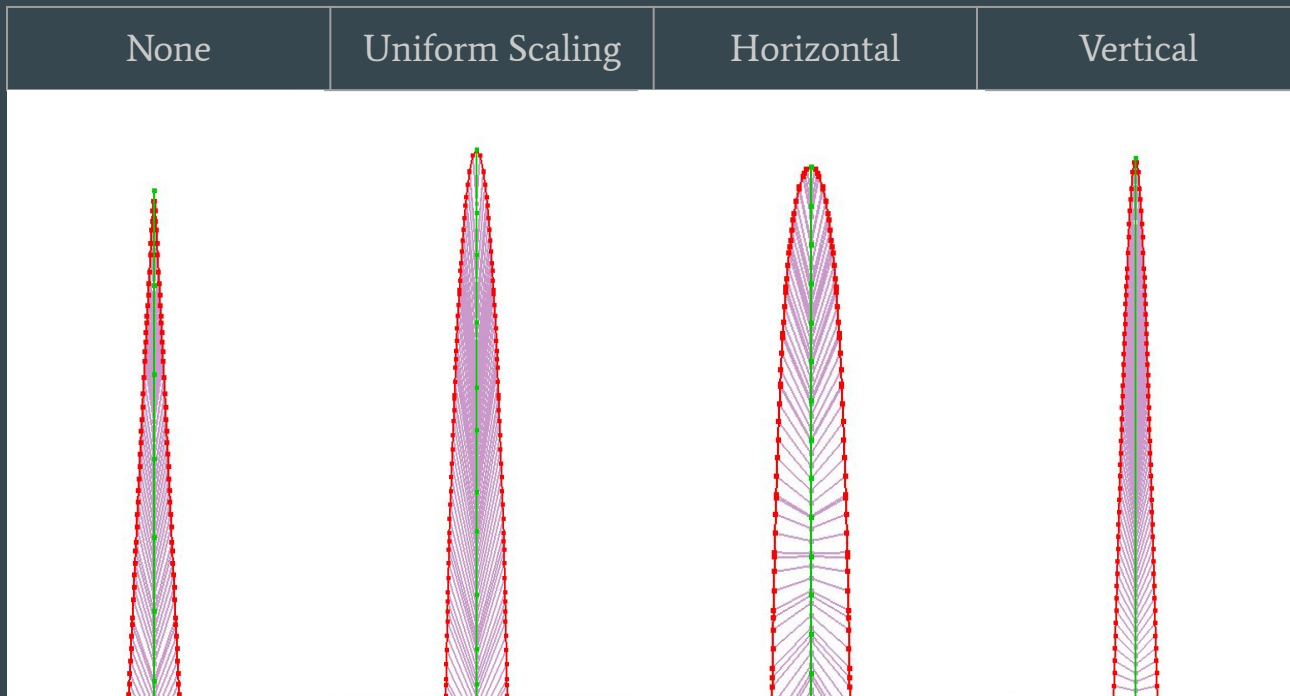
Different types of growth gradients are seen in biology as well, noted and studied in Nicotra *et al.* (2011)—*The evolution and functional significance of leaf shape in the angiosperms.*

x-axis: position on branch
y-axis: growth rate

Using branch scaling rate



Varying Parameter: Branch Scaling



non-uniform

Furthermore, modifying the branch scaling parameter can influence the tip form, determining whether it appears pointed or rounded. Even within the same species, such as *Alstonia congensis*, variation in tip shape is commonly observed.

Findings/Results: Comparison to Real Leaves

Alstonia congensis



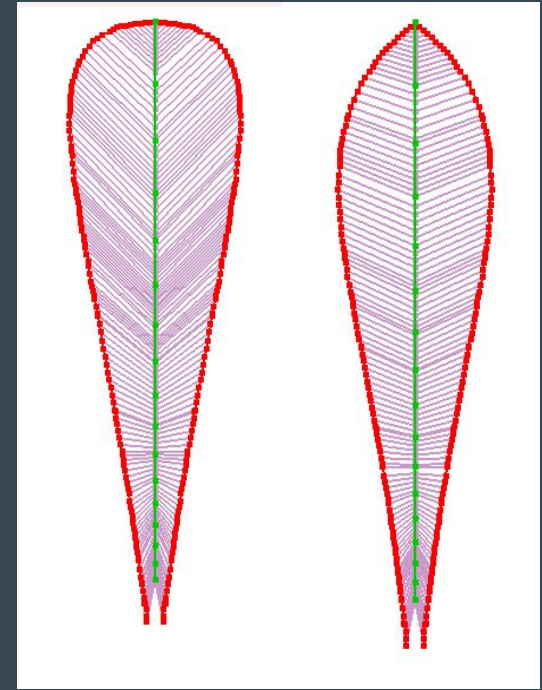
Fern, K. (n.d.). *Alstonia congensis* [Photograph]. In *Useful Tropical Plants* (Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported)



Fig. 59
Obovate leaf shape
Alstonia congensis
(Apocynaceae)



PlantNet contributors. (n.d.). *Alstonia congensis* [Photograph]. (Creative Commons Attribution-ShareAlike 4.0 International).

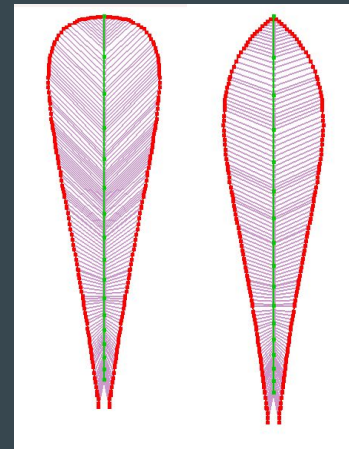


Horizontal branch
scaling

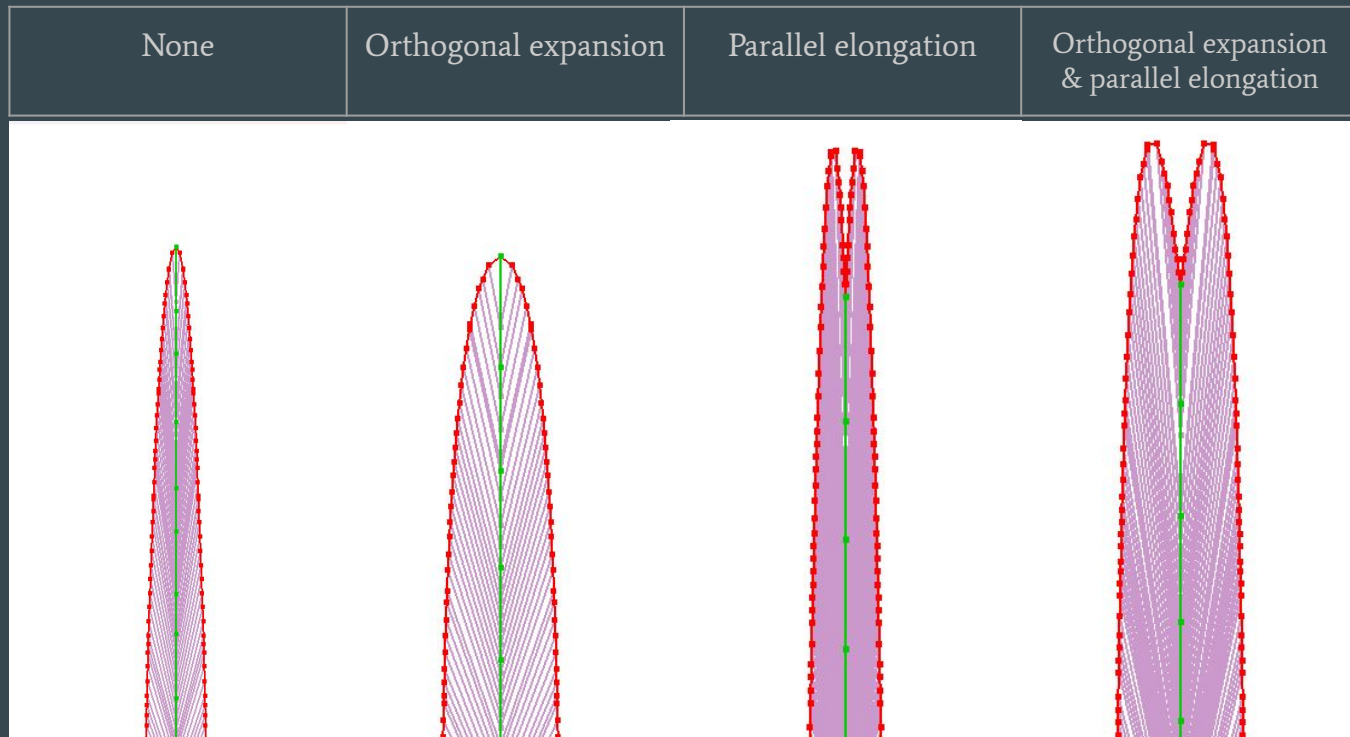
Uniform branch
scaling

Shape Analysis & Simulation Parameter: *Alstonia congestis*

- Leaf shape architecture:
 - Obovate to narrowly obovate
 - Obovate: ovate (egg-shaped) with narrower end at the base
 - Wider at the top than the base
- Simulation:
 - Branch growth (hierarchical structure):
 - Acropetal growth gradient
 - Local translation + local scaling + animated scaling = scaling + translation
 - Animated scaling is responsible for branch growth
 - Changes/deforms over time
 - Margin growth (from binding + additional non-hierarchical growth for margin only):
 - Branch scaling + translation: for expansional growth/elongation
 - Branch local position is updated with elongation
 - Orthogonal margin growth + acropetal branch growth: for obovate (wide top) shape
 - Different tip shapes are adjusted by branch scaling parameter



Varying Parameter: Margin Expansion & Elongation



Equivalent to BL expansion

Similar to branch scaling, margin expansion and elongation also contribute to shaping the leaf tip. Varying expansion and/or elongation can produce tips that are deeply indented or nearly flat. For example, *Magnolia officinalis* exhibits a range of indentation depths at the leaf tip.

- ★ Note that this only affects the margin points – this is not part of the hierarchical branching structure transformations

Findings/Results: Comparison to Real Leaves

Magnolia Officinalis

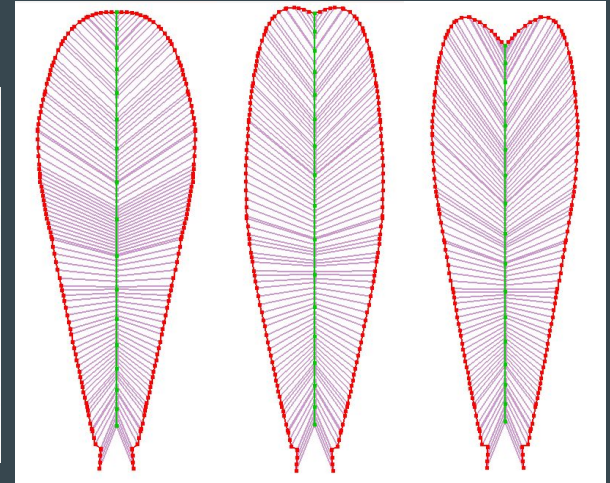
De Langhe, J.
(2014,
August 25).
Foliage on
Magnolia
officinalis var.
biloba at
Arboretum
Wespelaar
[Photograph].



de Spoelberch, P.
(2011,
September 15).
Leaves of *Magnolia*
officinalis arranged
in false whorls at
branch tips (Jiangxi
provenance,
Arboretum
Wespelaar)
[Photograph].



De Langhe, J. (2014, September 10). *Obovate lamina with glaucous-green lower surface, slightly emarginate apex, Magnolia officinalis* (Tregrehan Garden, UK) [Photograph].



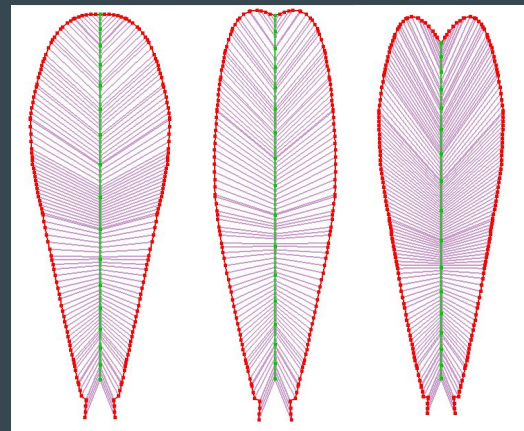
No BL
expansion

BL expansion of
rate 0.5

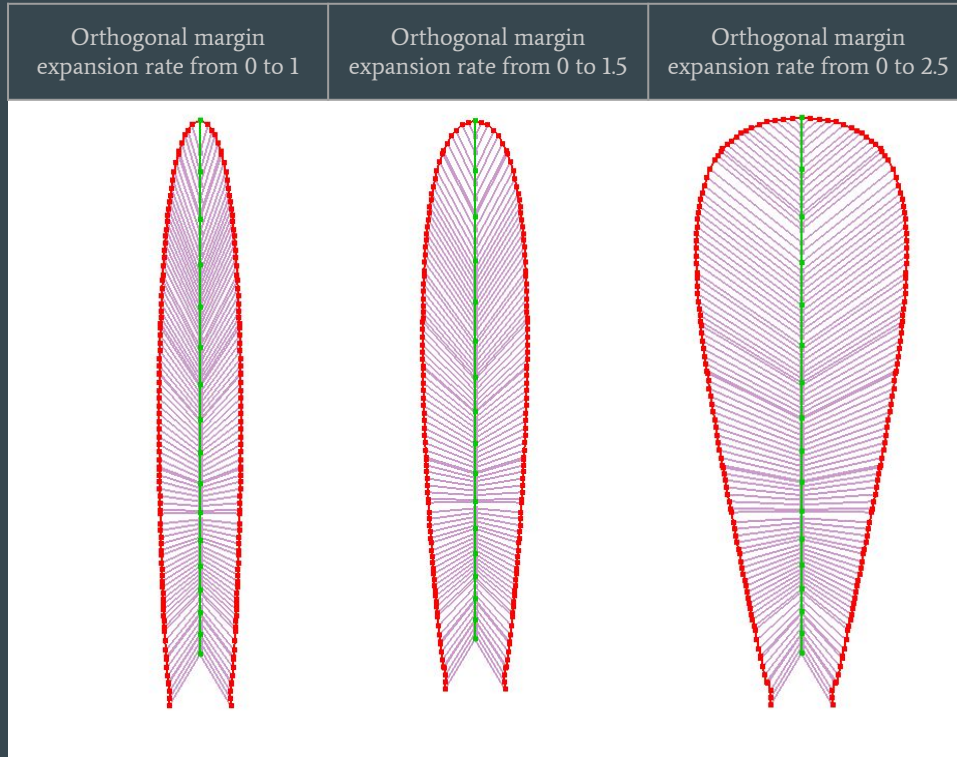
BL expansion of
rate 1.0

Shape Analysis & Simulation Parameter: *Magnolia Officinalis*

- Leaf shape architecture:
 - Broad, obovate (egg-shaped)
 - Some varieties may have notched or lobed tip
- Simulation:
 - Branch growth:
 - Bidirectional growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation + scaling for growth: for expansional growth
 - BL expansion: for lobed tips



Varying Parameter: Margin Expansion



Orthogonal margin expansion further helps in controlling the width of the leaf lamina/blade and the overall spatial area. Some leaves, such as *bamboo* or *willow leaves* are narrow, where was leaves like *banana leaves* or *Sonneratia Ovata* have a broader width.

→ With acropetal growth gradient

Findings/Results: Comparison to Real Leaves

Banana leaf

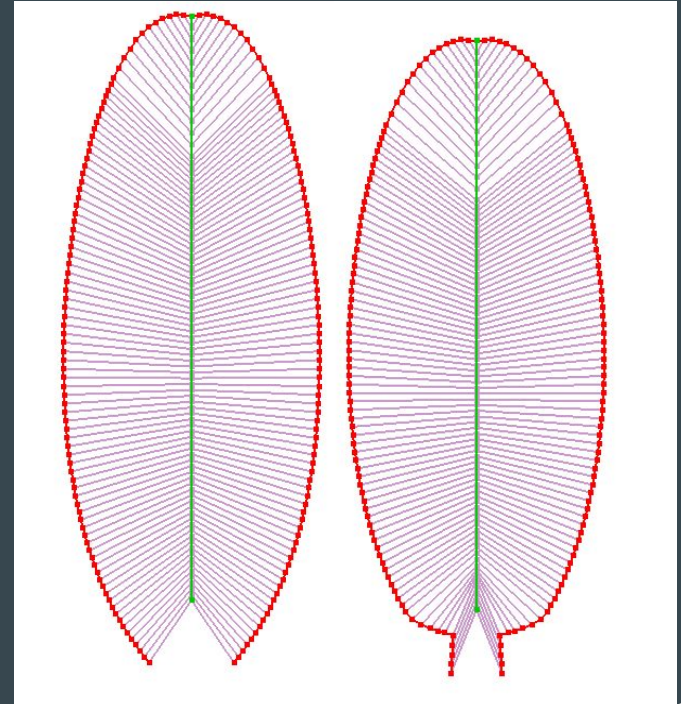
Reliance Retail. (n.d.). *Banana leaves* [Photograph].



shmar-4103. (n.d.). *Fresh banana leaves* [Photograph].



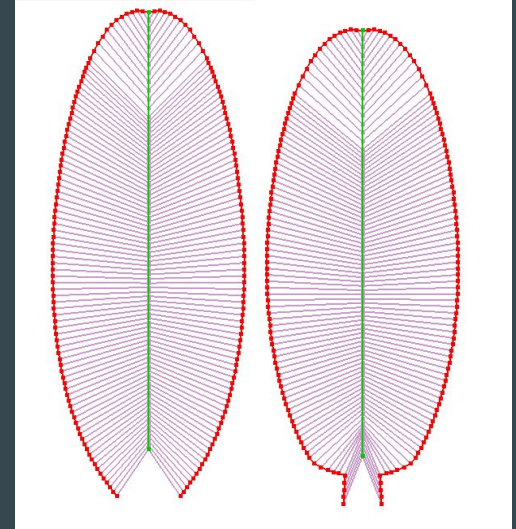
Bunches Direct. (n.d.). *Banana leaf* [Photograph].



Petiole

Shape Analysis & Simulation Parameter: Banana leaf

- Leaf shape architecture:
 - Elongated and paddle-shaped
 - Large, flat, wide spreading shape
 - Rounded edges, oblong leaf
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation + scaling: for expansional growth
 - Orthogonal expansion: for wide spreading shape
 - Small amount of parallel elongation: for flat tip

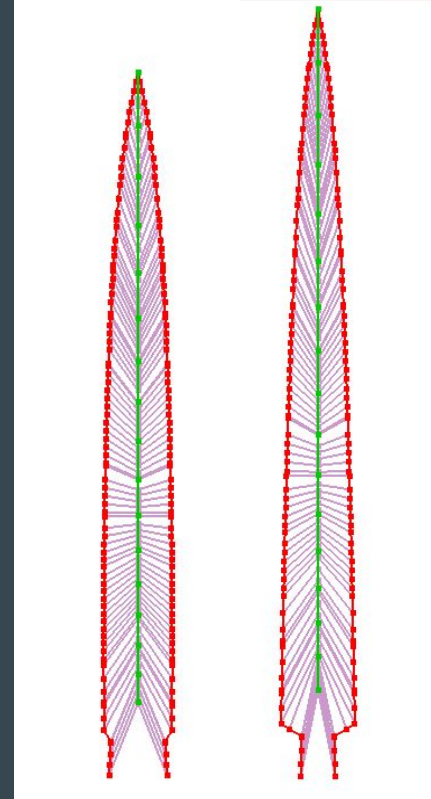


Findings/Results: Comparison to Real Leaves

Bamboo leaf

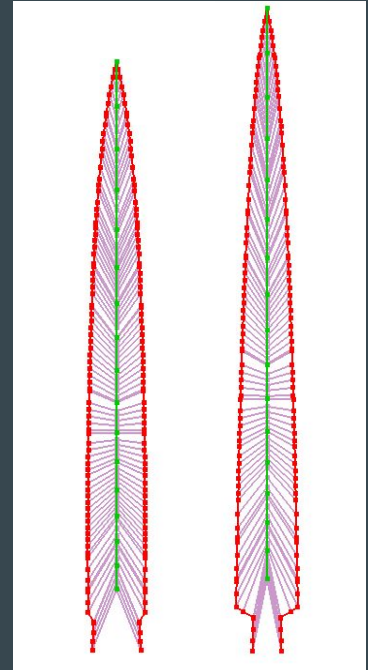


Various contributors. (n.d.). *Bamboo* [Photograph]. In *Wikipedia* (CC BY-SA 3.0).



Shape Analysis & Simulation Parameter: Bamboo leaf

- Leaf shape architecture:
 - Long and lanceolate
 - Lanceolate: ovate leaves that are long and narrow
 - Narrow and pointed at both ends
 - Taper to a point at the end
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation: for parallel growth/elongation
 - Scaling does not affect the margin points
 - Creates a narrow and pointed shape
 - Orthogonal expansion for giving spatial area to the leaf lamina
 - Branch growth rate must be higher than orthogonal margin growth rate
 - Preserves the narrow shape



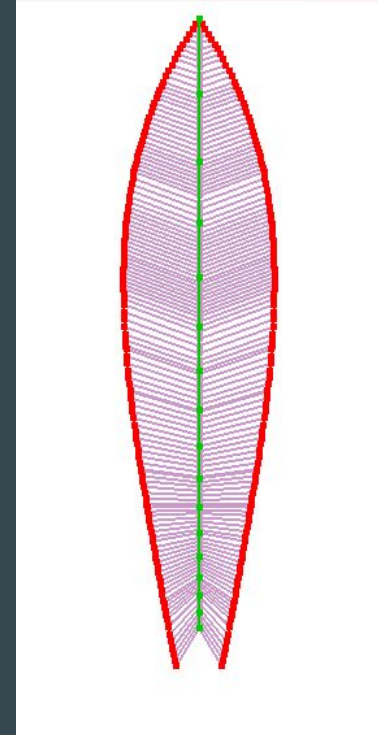
Findings/Results: Comparison to Real Leaves

Purple leaf plum/Cherry plum



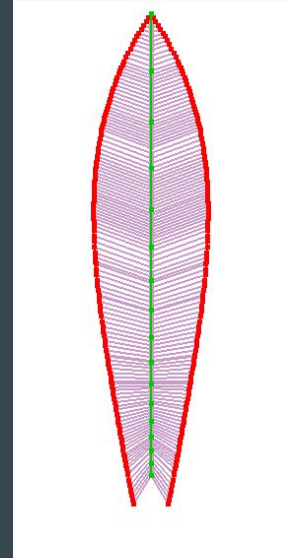
Ancient Gardener. (2011, May). *Purple-leaf plum trees affected by Japanese beetles* [Photograph]. Ancient Gardener Blog.

Figure 1. Photo by author.



Shape Analysis & Simulation Parameter: Purple leaf plum

- Leaf shape architecture:
 - Oval shape (vase-shape) with pointed tip
 - Vase shape: wider, more open form at the top that tends to spread outwards
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation: for parallel growth/elongation
 - Scaling does not affect the margin points
 - There is only parallel growth
 - Creates a sharp, pointed tip
 - Orthogonal expansion + acropetal branch growth: for obovate (wide top) shape



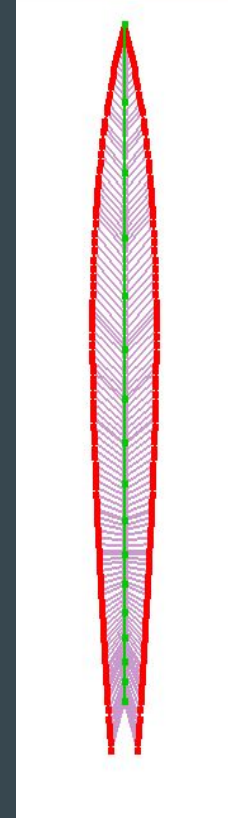
Findings/Results: Comparison to Real Leaves

Oleander leaf



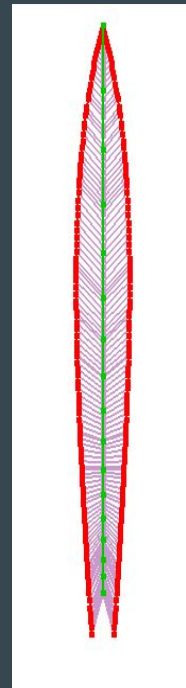
CAB International. (n.d.). *Nerium oleander* [Photograph]. CABI Compendium (© CAB International; non-commercial academic use).

Vanderpoel, J. H. (n.d.). *Nerium oleander* [Photograph]. Backyard Nature.



Shape Analysis & Simulation Parameter: Oleander leaf

- Leaf shape architecture:
 - Linear-lanceolate shape
 - Long and narrow
 - Pointed tip and smooth margin
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation
 - Orthogonal expansion
- Is there any difference from the bamboo leaves?
 - Unlike bamboo leaves, the orthogonal expansion rate is either the same or higher than translation branch growth rate
 - Oleander leaves are slightly wider near the tip compared to bamboo leaves



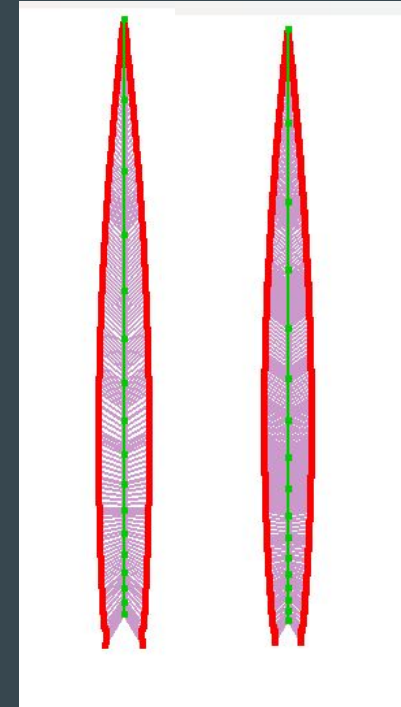
Findings/Results: Comparison to Real Leaves

Willow leaf



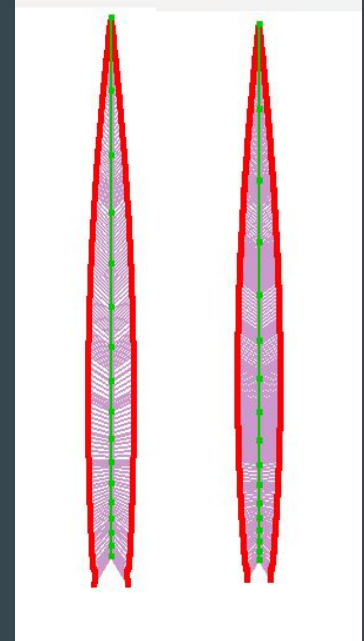
Gilman, E. F., & Warfield, C. (n.d.).
Salix babylonica [Illustration].
UF/IFAS Extension.

Government of Ontario. (2025, August 14). *Salix amygdaloides*
(Peachleaf willow) [Photograph].



Shape Analysis & Simulation Parameter: Willow leaf

- Leaf shape architecture:
 - Lanceolate (spear-shaped)
 - Long and slender with pointed tip
 - Some may have slender, tail-like extension in the tip
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation
 - Orthogonal expansion
 - Small to preserve the elongated, slender shape but still give spatial area to the lamina (blade)



Findings/Results: Comparison to Real Leaves

Diamond Willow leaf



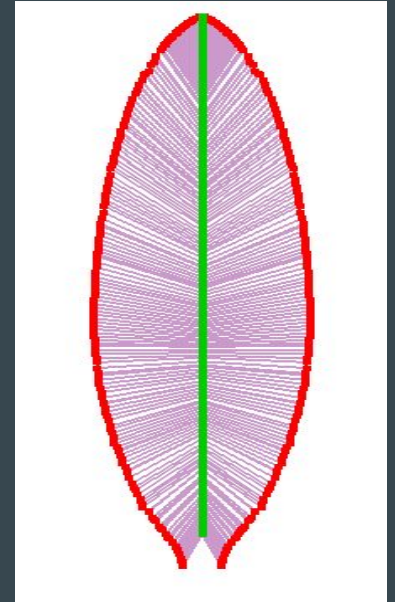
USDA-NRCS PLANTS Database / Britton & Brown. (1913). *Salix eriocephala* [Illustration]. (Public domain, via Wikipedia).



Dziuk, P. M. (2018). *Salix planifolia* (Diamond-leaf Willow) [Photograph]. Minnesota Wildflowers. Accessed 14 August 2025; also accessed 19 August 2025.

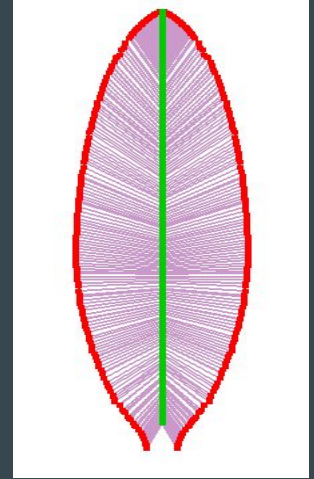


Halley, T. (n.d.). *Salix planifolia* (Plane-leaf Willow) [Photograph]. Science & Nature: Cascadia Wildflowers. Accessed 22 August 2025, from <http://science.halleyhosting.com/nature/cascade/shrubs/salix/planifolia.html>



Shape Analysis & Simulation Parameter: Diamond willow leaf

- Leaf shape architecture:
 - Lanceolate to elliptic shape
 - Narrowly urn-shaped form
 - Broader in the middle and tapering towards both ends
- Simulation:
 - Branch growth:
 - Bidirectional growth gradient
 - Acropetal growth gradient (growth gets higher towards tip) from base to midpoint
 - Basipetal growth gradient (growth gets lower towards tip) from midpoint to tip
 - Translation + scaling for growth
 - Margin growth:
 - Translation + scaling: for expansional growth
 - Orthogonal expansion + diffuse branch growth: for mid-axis expansion



Findings/Results: Comparison to Real Leaves

Diamond-leaf Pittosporum

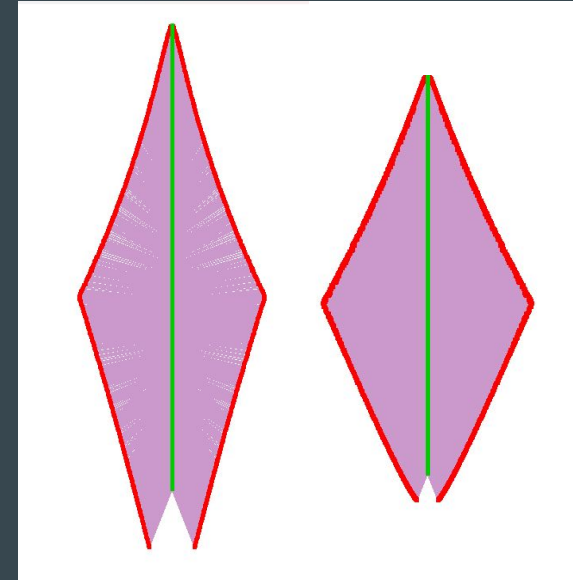


Royal Botanic Gardens and Domain Trust. (n.d.).
Auranticarpa rhombifolia (A.Cunn. ex Hook.) L.W.Cayzer, Crisp & I.Telford. In
PlantNET (The NSW Plant Information Network System).
Retrieved August 21, 2025, from
<https://plantnet.rbgsyd.nsw.gov.au/cgi-bin/NSWfl.pl?page=nswfl&lvl=sp&name=Auranticarpa-rhombifolia>



Palmbob. (2010, November 16). Fall, Southern California [Photograph]. Dave's Garden.
<https://davesgarden.com/guides/pff/showimage/294726>

Whyte, R. (n.d.). Flower buds [Photograph].
Sown.
<https://sown.com.au/auranticarpa-rhombifolia-pittosporaceae-diamond-leaved-pittosporum/>

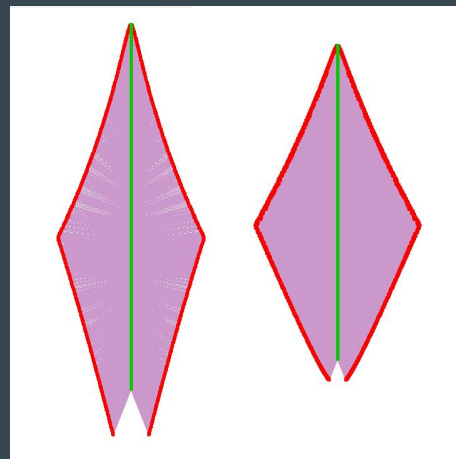


Lower orthogonal
expansion rate

Higher orthogonal
expansion rate

Shape Analysis & Simulation Parameter: Diamond-leaf pittosporum

- Leaf shape architecture:
 - Diamond or rhomboid shape
 - Toothed in top half, untoothed in bottom half
- Simulation:
 - Branch growth:
 - Bidirectional growth gradient
 - Acropetal growth gradient (growth gets higher towards tip) from base to midpoint
 - Basipetal growth gradient (growth gets lower towards tip) from midpoint to tip
 - Translation + scaling for growth
 - Margin growth:
 - Translation + scaling: for expansional growth
 - Orthogonal expansion + diffuse branch growth: for mid-axis expansion



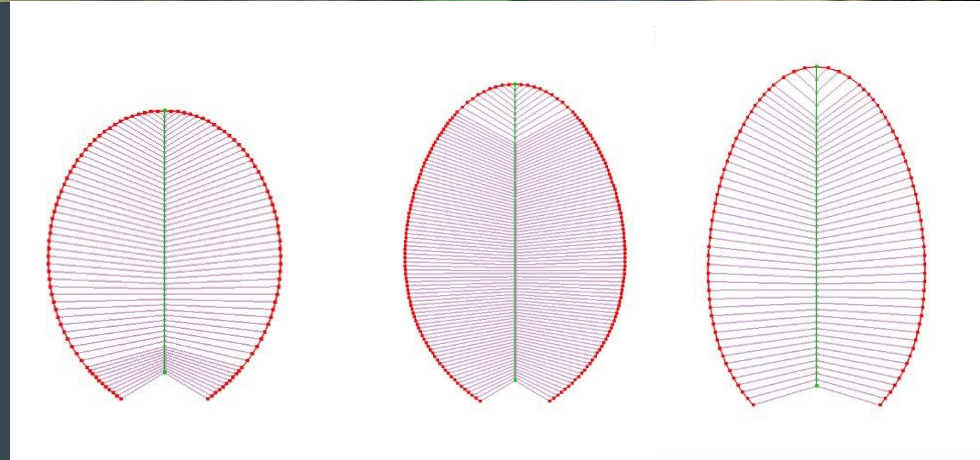
Findings/Results: Comparison to Real Leaves

Sonneratia Ovata:

NParks Flora & Fauna Web
contributors. (2025, August 14).
Sonneratia ovata [Photograph].
Flora & Fauna Web.



NParks Flora & Fauna Web
contributors. (2025, August 14).
Sonneratia ovata [Photograph].
Flora & Fauna Web.



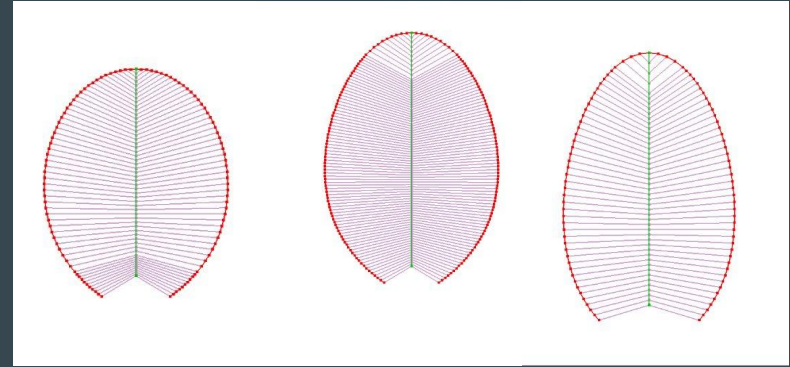
Animated scaling rate is 0

Animated scaling rate is
0.3

Animated scaling rate is
0.5

Shape Analysis & Simulation Parameter: Sonneratia Ovata

- Leaf shape architecture:
 - Ovate or suborbicular shape
 - Ovate: widest part is towards the base
 - Suborbicular: nearly circular
 - Broadly egg-shaped to nearly circular
 - Rounded base and rounded apex
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling
 - **Animate scaling rate is 0 or low**
 - Almost no growth involved
 - Branch position stays almost the same
 - Margin growth:
 - Orthogonal expansion
 - No elongation/parallel growth
 - Translation: since there is no branch growth, there is no elongation
 - Branch length stays the same



Findings/Results: Comparison to Real Leaves

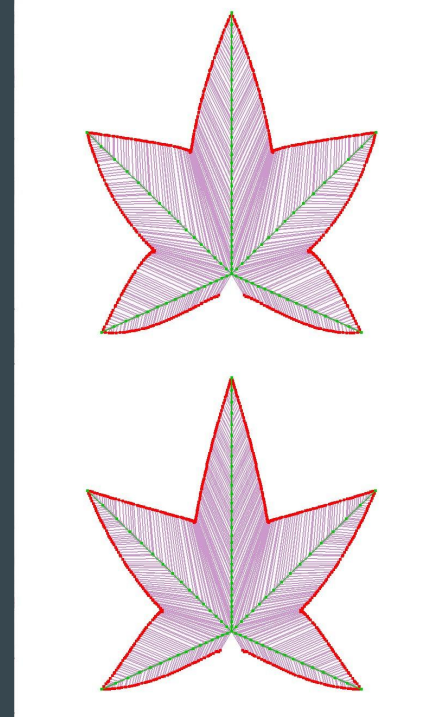
Sweetgum

Doe, J. (n.d.). *California Poppy*
(*Eschscholzia californica*)
[Photograph]. iNaturalist.
(CC-BY-NC).

https://www.inaturalist.org/guide_taxa/526268



Oklahoma State University Extension. (n.d.).
Liquidambar styraciflua (Sweetgum) [Photograph].
Plant Identification Profiles.

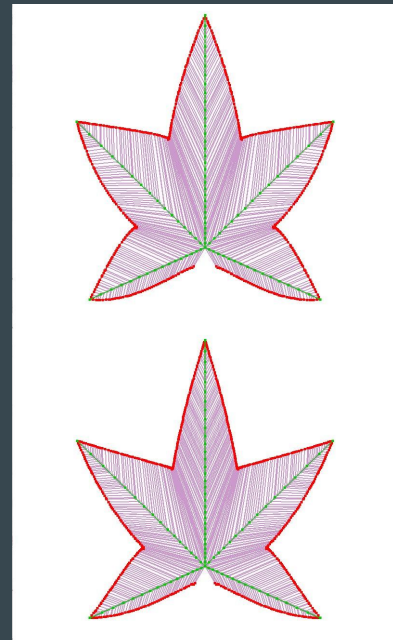


Orthogonal expansion
rate is from 0 to 1

Orthogonal expansion
rate is from 0 to 0.8

Shape Analysis & Simulation Parameter: Sweetgum

- Leaf shape architecture:
 - Distinctive star-shape
 - Typically with 5 (more common) to 7 pointed lobes
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation: for vertical elongation
 - Allows for pointed tip
 - Orthogonal expansion
 - Orthogonal expansion rate must be handled with care to prevent margin overlap between lobes



Findings/Results: Comparison to Real Leaves

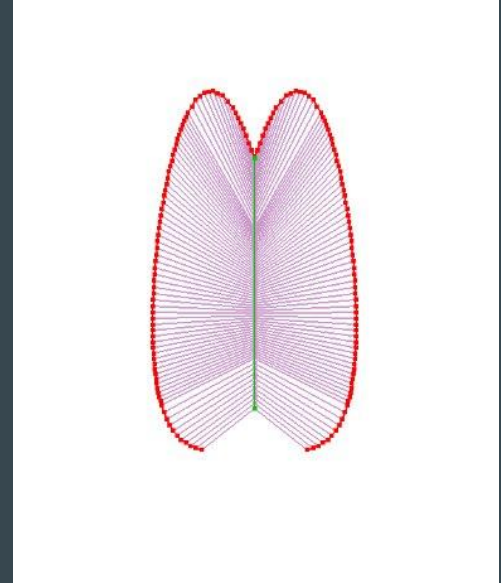
Orchid Tree Leaf



Orchid Tree (Bauhinia variegata) [Description]. In *South Florida Trees: A Guide to the Trees of South Florida and the Keys* (2025).

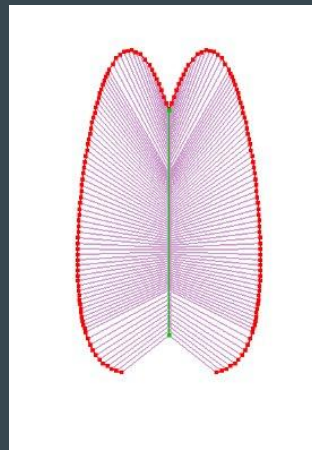


Bauhinia x blakeana (Hong Kong Orchid Tree) [Description]. (2025, August 14). In *Gardenia*. Retrieved August 14, 2025.



Shape Analysis & Simulation Parameter: Orchid tree leaf

- Leaf shape architecture:
 - Rounded with bilobed or double-lobed ends
 - Divided into two distinct lobes, separated by cleaf or indentation at the tip
 - Resemble a heart shape or butterfly wing shape
 - Wider than long
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Local translation + local scaling + orthogonal animated scaling for growth
 - Margin growth:
 - BL expansion: for heart-shaped tip
 - Translation: since there is no parallel branch scaling, there is no branch growth
 - Branch length stays the same



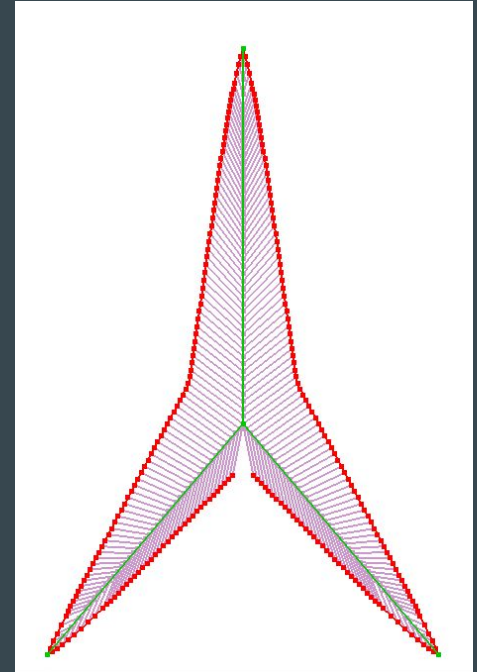
Findings/Results: Comparison to Real Leaves

Arrowhead



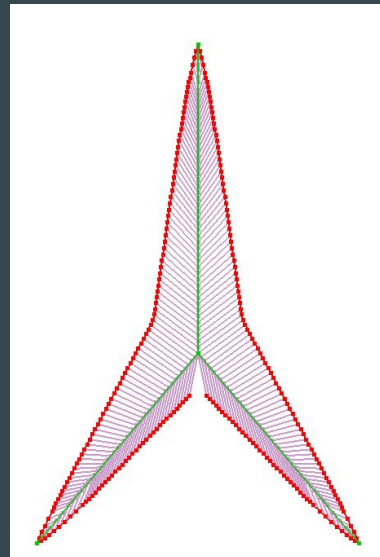
Maxerová, J. (2010, July). *Sagittaria sagittifolia* leaf, Ohře river, Czech Republic [Photograph]. Wikimedia Commons (CC-BY-SA 3.0).

Sovík, Z., & Tračák. (2007, July 24). *Bryomech specimen* [Photograph]. CHKO Poodří, Polanka nad Odrou, Czechia. Via BioLib.cz.



Shape Analysis & Simulation Parameter: Arrowhead

- Leaf shape architecture:
 - Sagittate (arrowhead) shape
 - Pointed tip and two basal lobes that extend backward
- Simulation:
 - Branch growth:
 - Acropetal growth gradient
 - Translation + scaling for growth
 - Margin growth:
 - Translation: for parallel elongation
 - Creates elongated, pointed lobes



Findings/Results: Comparison to Real Leaves

Tulip Tree Leaf



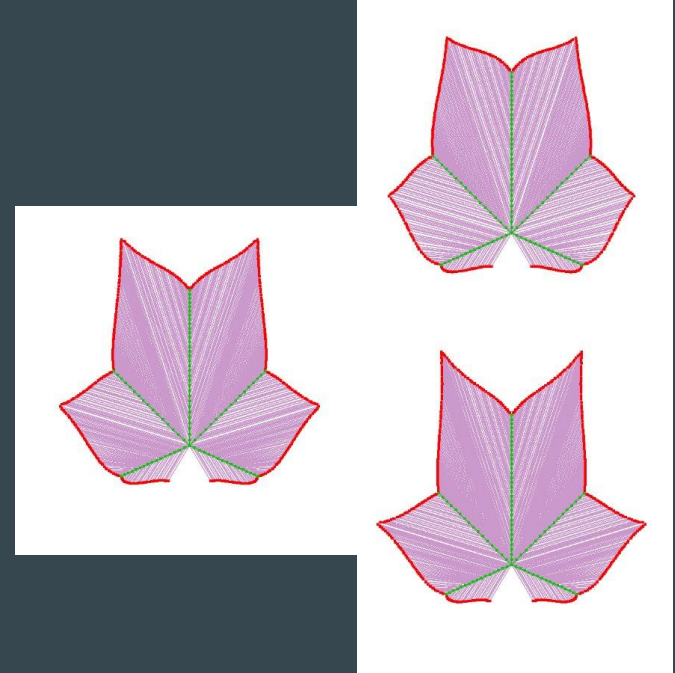
Ramsey, D. (2007, May 27). Photograph of topside of a Tulip Tree (*Liriodendron tulipifera*) leaf. Wikimedia Commons (GFDL 1.2).



Various contributors. (2025, August 11). *Liriodendron foliage and flowers* [Illustration]. Wikipedia (CC BY-SA 4.0).

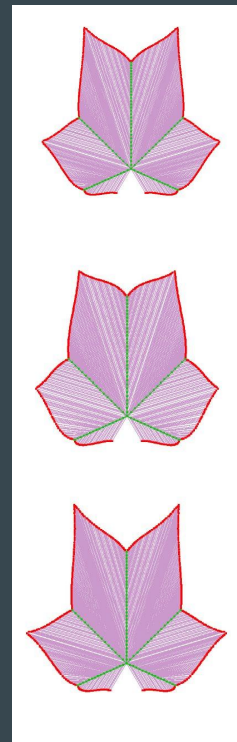


Smithsonian Gardens. (2025, August 14). *Liriodendron tulipifera* (Tulip Tree – ...) [Description]. Plant Explorer, Smithsonian Gardens.



Shape Analysis & Simulation Parameter: Tulip tree leaf

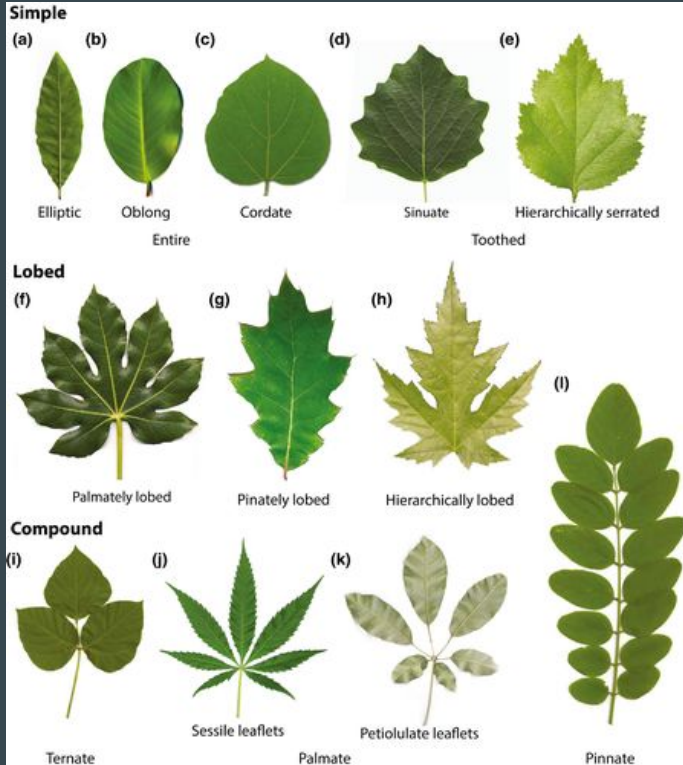
- Leaf shape architecture:
 - Distinct four-lobed square or truncate shape
 - Typically broad, flat top and four pointed lobes along the bottom
 - Heart-shaped or truncate at the base
- Simulation:
 - Branch growth:
 - Basipetal growth gradient (growth at base is higher and gets lower towards tip)
 - Translation + scaling for growth
 - Margin growth:
 - Translation + scaling: for expansional growth
 - Since growth rate is highest in the base/root, margin points associated with the base grows the fastest



Conclusion: Findings & Observations

- With the results, we can conclude that skinning was a successful approach that allowed us to generate simple leaf shapes **without** the help of additional ad-hoc methods
 - Removes the need to use biologically unacceptable/unexplainable methods
- With a careful binding of margin points to the branching structure and adjusting the parameters accordingly (such as transformation type, growth rate, shape of branching structure, placement of initial margin, skinning type, etc), it is possible to generate diverse shapes of leaves that we see in nature
- Moreover, we are able to capture how leaves within species or leaves within related species have slightly different shapes by only varying one parameter
 - Simple transition between related shapes, which is biologically accurate
- Transformations, particularly scaling, must be handled with care as different operation orders and growth gradient will drastically change the resulting shape.
 - Growth is accumulated exponentially, hence linearly blending them might introduce artifacts that must be handled carefully

Conclusion: Limitations

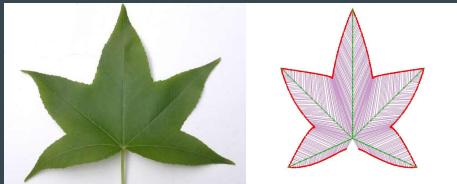


- The current model can generate simple and lobed leaves, but compound and hierarchical are hard to generate
 - Hierarchical vein structure: larger veins branch into smaller veins
- Dynamic addition of new veins/axis is possible but has limitations
 - Margin points must be rebound properly to preserve the smooth curve
 - Dynamic addition of veins is important to accurately simulate development, especially in simulating growth for simple and lobed leaves
 - New veins start appearing as leaves grow

Conclusion: Local Future Plans

- Hence, we plan to proceed further with the project and expand the model to be able to generate more complex shapes as well:
 - Be able to generate compound and hierarchical leaves
 - Be able to dynamically add new veins/axis and preserve contour shape
 - Be able to better control details such as indentations and tip shapes
 - Be able to control patterning of margin
 - To create variety of shapes along the leaf's edge such as serrations, lobes, or leaflets

Oklahoma State
University
Extension. (n.d.).
*Liquidambar
styraciflua*
(Sweetgum)
[Photograph].
Plant Identification
Profiles.



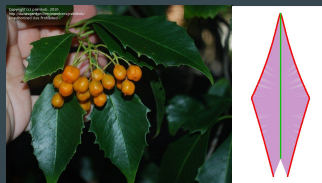
For sweetgum, the tips taper to a long, narrow point where the margin curves inwards as they converge. But the indentations are more sharp and does not curve. These differences **within** a leaf are hard to capture.



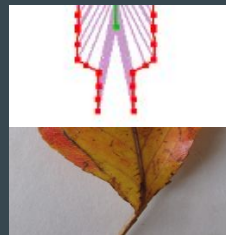
Even though indentations are controllable to a certain extent, we are unable to create a deeply rounded indentation such as the figure here

Figure adapted from Runions et al. (2017),
Figure 7. [New
Phytologist, 216(2), 401–418.

Whyte, R. (n.d.).
Flower buds
[Photograph]. Sown.
<https://sown.com.au/a-uranticarpa-rhombifolia-pittosporaceae-diamond-leaved-pittosporum/>



For Diamond-leaf Pittosporum, real leaves have **serration** (margin is notched with sharp, forward-pointing teeth), which is hard to capture in the simulation

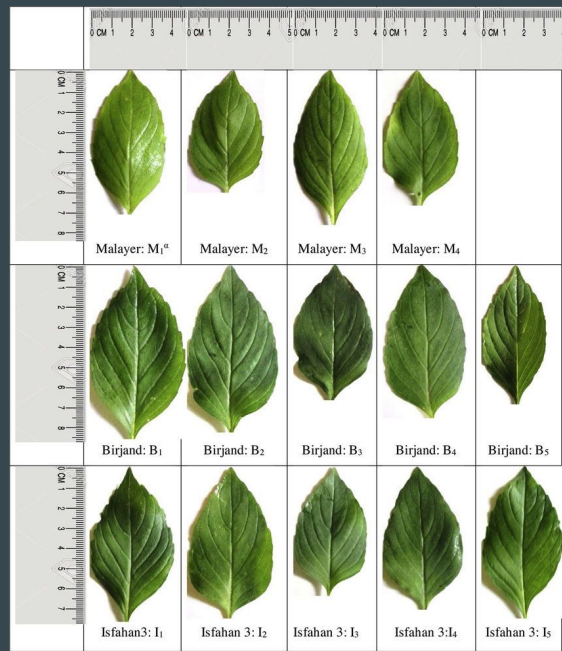


Currently, regions near the petiole tends to make a sharp angle, limiting from achieving shapes that have a rounder margin near the petiole

Figure adapted from
https://en.wikipedia.org/wiki/Petiole_%28botany%29

Conclusion: Global Future Plans

- We also want to be able to quantify growth: infer growth parameters from a given contour (reverse engineer)
 - Parameter reuse: Apply extracted parameters to generate similar or variant shapes
 - Design flexibility: Modify or interpolate between parameters to explore new morphologies
 - Model validation: Compare inferred parameters with known biological data
- On the computer graphics point of view...
 - Apply texture to make leaves more realistic
 - Tweak the parameters of a contour to get a cluster of varying leaf shapes within species
 - Mirroring how leaf shapes are slightly different even within species
 - During and after simulation, be able to vary parameters to reduce symmetry
 - E.g. one lobe is smaller than the other
 - Extend to surfaces
 - Model the entire plant, not just leaf



Putievsky, E., Ravid, U., & Dudai, N. (1999). Selection of basil (*Ocimum basilicum* L.) full-sib families from diverse landraces. *Journal of Herbs, Spices & Medicinal Plants*, 6(2), 19–26. https://doi.org/10.1300/J044v06n02_03

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Ellis, B., Daly, D. C., Hickey, L. J., Johnson, K. R., Mitchell, J. D., Wilf, P., & Wing, S. L. (2009). *Manual of leaf architecture*. Cornell University Press.

Jacobson, A., Deng, Z., Kavan, L., & Lewis, J. P. (2014). *Skinning: Real-time shape deformation*. In *ACM SIGGRAPH 2014 Courses*.

Jacobson, A., & Sorkine, O. (2011). *Stretchable and twistable bones for skeletal shape deformation*. *ACM Transactions on Graphics (TOG)*, **30**(6), 165. <https://doi.org/10.1145/2070781.2024160>

Runions, A., Tsiantis, M., & Prusinkiewicz, P. (2017). A common developmental program can produce diverse leaf shapes. *New Phytologist*, **216**(2), 401–418. <https://doi.org/10.1111/nph.14449>